Associative algebra

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Abstract. The notes correspond to the master course **Associative Algebra** of the Vrije Universiteit Brussel, Faculty of Sciences, Department of Mathematics and Data Sciences.

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Introduction

The notes correspond to the master course **Associative Algebra** of the Vrije Universiteit Brussel, Faculty of Sciences, Department of Mathematics and Data Sciences. The course is divided into twelve two-hour lectures.

The reader should have a solid understanding of an undergraduate-level abstract algebra course. For reference, you can review my notes for the VUB courses: Group Theory and Ring and Module Theory.

The content presented here draws heavily from [**5**], [**10**], and [**20**]. Additionally, I have followed the outstanding blog on abstract algebra by Yaghoub Sharifi.

The notes include many exercises, some with full detailed solutions. Mandatory exercises have a green background, while optional ones (bonus exercises) have a yellow background.

The notes also include some additional comments. While these are entirely optional, I hope they offer further insight. They are highlighted with a pink background.

The notes include Magma code, which we use to verify examples and offer alternative solutions to certain exercises. Magma [**4**] is a powerful software tool designed for working with algebraic structures. There is a free online version of Magma available.

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§ 1. Semisimple algebras

We will study finite-dimensional semisimple algebras. The main goal of the first two lectures is to prove the Artin–Wedderburn theorem.

1.1. Definition. An **algebra** (over the field K) is a vector space (over K) with an associative multiplication $A \times A \to A$ such that

$$a(\lambda b + \mu c) = \lambda(ab) + \mu(ac) \quad \text{and} \quad (\lambda a + \mu b)c = \lambda(ac) + \mu(bc)$$

for all $a, b, c \in A$ and $\lambda, \mu \in K$, and contains an element $1_A \in A$ such that

$$1_A a = a 1_A = a$$

for all $a \in A$.

Note that a ring A is an algebra over K if and only if there is a ring homomorphism $\iota \colon K \to Z(A)$, where $Z(A) = \{a \in A : ab = ba \text{ for all } b \in A\}$ is the **center** of A, such that $\iota(1_K) = 1_A$.

1.2. Definition. An algebra A is **commutative** if $ab = ba$ for all $a, b \in A$.

The **dimension** of an algebra A is the dimension of A as a vector space. This is why we want to consider algebras, as they are a linear version of rings. Often, our arguments will use the dimension of the underlying vector space.

1.3. Example. The field $\mathbb{R}$ is a real algebra and $\mathbb{C}$ is a complex algebra. Moreover, $\mathbb{C}$ is a real algebra.

Any field K is an algebra over K.

1.4. Example. If K is a field, then $K[X]$ is an algebra over K.

Similarly, the polynomial ring $K[X, Y]$ and the ring $K[\![X]\!]$ of power series are examples of algebras over K.

1.5. Example. Let $n \geq 1$. If A is an algebra, then $M_n(A)$ is an algebra.

1.6. Example. The set of continuous maps $[0, 1] \to \mathbb{R}$ is a real algebra with the usual pointwise operations $(f + g)(x) = f(x) + g(x)$ and $(fg)(x) = f(x)g(x)$.

1.7. Example. Let $n \in \mathbb{Z}_{>0}$. Then $K[X]/(X^n)$ is a finite-dimensional K-algebra. It is the **truncated polynomial algebra**.

1.8. Example. Let G be a finite group. The vector space $\mathbb{C}[G]$ with basis $\{g : g \in G\}$ is an algebra with multiplication

$$\left(\sum_{g \in G} \lambda_g g\right)\left(\sum_{h \in G} \mu_h h\right) = \sum_{g,h \in G} \lambda_g \mu_h (gh).$$

Note that $\dim \mathbb{C}[G] = |G|$ and $\mathbb{C}[G]$ is commutative if and only if G is abelian. This is the **complex group algebra** of G.

If G is an infinite group, the complex group algebra $\mathbb{C}[G]$ is defined as the set of finite linear combinations of elements of G with the usual operations.

1.9. Definition. Let K be a field and A and B be K-algebras. An algebra **homomorphism** is a unital ring homomorphism $f \colon A \to B$ that is also a K-linear map.

The complex conjugation map $\mathbb{C} \to \mathbb{C}$, $z \mapsto \overline{z}$, is a ring homomorphism that is not an algebra homomorphism over $\mathbb{C}$.

1.10. Exercise. Let G be a non-trivial finite group. Then $\mathbb{C}[G]$ has zero divisors.

If A is an algebra, then $\mathcal{U}(A)$ is the group of units of A.

1.11. Exercise. Let A be a K-algebra and G be a finite group. If $f\colon G \to \mathcal{U}(A)$ is a group homomorphism, then there exists a unique algebra homomorphism $\varphi\colon K[G] \to A$ such that $\varphi|_G = f$.

1.12. Definition. An **ideal** of an algebra is an ideal of the underlying ring.

Similarly, one defines left and right ideals of an algebra.

If A is an algebra, then every left ideal of the ring A is a vector space. Indeed, if I is a left ideal of A and $\lambda \in K$ and $x \in I$, then

$$\lambda x = \lambda(1_A x) = (\lambda 1_A)x.$$

Since $\lambda 1_A \in A$, it follows that $\lambda I = (\lambda 1_A)I \subseteq I$. Similarly, every right ideal of the ring A is a vector space.

If A is an algebra and I is an ideal of A, then the quotient ring A/I has a unique algebra structure such that the canonical map $A \to A/I$, $a \mapsto a + I$, is a surjective algebra homomorphism with kernel I.

1.13. Definition. Let A be an algebra over the field K. An element $a \in A$ is **algebraic** over K if there exists a non-zero polynomial $f \in K[X]$ such that $f(a) = 0$.

If every element of A is algebraic, then A is said to be **algebraic**.

$\mathbb{R}$ is not algebraic as a $\mathbb{Q}$-algebra. A famous theorem of Lindemann proves that π is not algebraic over $\mathbb{Q}$. Every element of $\mathbb{R}$, regarded as an $\mathbb{R}$-algebra, is algebraic over $\mathbb{R}$.

1.14. Proposition. *Every finite-dimensional algebra is algebraic.*

Proof. Let A be an algebra with $\dim A = n$ and let $a \in A$. Since the $n + 1$ elements $1, a, a^2, \dots, a^n$ are linearly dependent, there exists a non-zero polynomial $f \in K[X]$ such that $f(a) = 0$. $\square$

Throughout Lectures 1 and 2, all modules are unital left modules unless otherwise stated.

1.15. Definition. A **module** over an algebra A is a unital left module over the ring A.

Similarly, one defines **submodules** of A-modules.

1.16. Definition. Let A be a K-algebra. A **homomorphism** of A-modules $f\colon M \to N$ is a K-linear map such that $f(a \cdot m) = a \cdot f(m)$ for all $a \in A$ and $m \in M$.

It is a straightforward exercise to prove the isomorphism theorems.

Let A be a finite-dimensional K-algebra. If M is a module over the ring A, then M is a vector space with

$$\lambda m = (\lambda 1_A) \cdot m,$$

where $\lambda \in K$ and $m \in M$. Moreover, M is finitely generated if and only if M is finite-dimensional.

1.17. Example. If M is a module over a finite-dimensional K-algebra A, one defines $\operatorname{End}_A(M)$ as the set of A-module homomorphisms $M \to M$. The set $\operatorname{End}_A(M)$ is indeed a K-algebra with

$$(f+g)(m) = f(m) + g(m), \quad (\lambda f)(m) = \lambda f(m) \quad \text{and} \quad (fg)(m) = f(g(m))$$

for all $f, g \in \operatorname{End}_A(M)$, $\lambda \in K$ and $m \in M$.

1.18. Example. An algebra A is a module over A with left multiplication, that is

$$a \cdot b = ab, \quad a, b \in A.$$

This module is the (left) **regular representation** of A and it will be denoted by ${}_AA$.

1.19. Definition. Let A be an algebra and M be a module over A. Then M is **simple** if $M \neq \{0\}$ and $\{0\}$ and M are the only submodules of M.

1.20. Definition. Let A be a finite-dimensional algebra and M be a finite-dimensional module over A. Then M is **semisimple** if M is a direct sum of finitely many simple submodules.

By definition, the zero module is semisimple. Moreover, any finite direct sum of semisimple modules is semisimple.

1.21. Lemma (Schur). *Let A be an algebra. If S and T are simple modules and $f\colon S \to T$ is a non-zero A-module homomorphism, then f is an isomorphism.*

Proof. Since $f \neq 0$, $\ker f$ is a proper submodule of S. Since S is simple, it follows that $\ker f = \{0\}$. Similarly, $f(S)$ is a non-zero submodule of T and hence $f(S) = T$, as T is simple. □

1.22. Proposition. *If A is a finite-dimensional algebra and S is a simple A-module, then S is finite-dimensional.*

Proof. Let $s \in S \setminus \{0\}$. Since S is simple, $\varphi\colon A \to S$, $a \mapsto a \cdot s$, is a surjective module homomorphism. In particular, by the first isomorphism theorem, $A/\ker\varphi \simeq S$ and hence

$$\dim_K S = \dim_K(A/\ker\varphi) \leq \dim_K A.$$ □

1.23. Proposition. *Let A be a finite-dimensional algebra and M a finite-dimensional A-module. The following statements are equivalent:*

1) *M is semisimple.*
2) *$M = \sum_{i=1}^{k} S_i$, where each S_i is a simple submodule of M.*
3) *If S is a submodule of M, then there is a submodule T of M such that $M = S \oplus T$.*

Proof. We first prove that 2) $\implies$ 3). If $N = \{0\}$, take $T = M$. Let $N \neq \{0\}$ be a submodule of M. Since $N \neq \{0\}$ and $\dim M < \infty$, there exists a submodule T of M of maximal dimension such that $N \cap T = \{0\}$. If $S_i \subseteq N \oplus T$ for all $i \in \{1, \dots, k\}$, then, as M is the sum of the S_i, it follows that $M = N \oplus T$. If, however, there exists $i \in \{1, \dots, k\}$ such that $S_i \not\subseteq N \oplus T$, then $S_i \cap (N \oplus T) \subseteq S_i$. Since the module S_i is simple, it follows that $S_i \cap (N \oplus T) = \{0\}$. Thus $N \cap (S_i \oplus T) = \{0\}$, a contradiction to the maximality of $\dim T$.

The implication 1) $\implies$ 2) is trivial.

Finally, we prove that 3) $\implies$ 1). We proceed by induction on $\dim M$. The result is clear if $\dim M \leq 1$. Assume that $\dim M \geq 2$ and let S be a non-zero submodule of M of minimal dimension. In particular, S is simple. By assumption, there exists a submodule T

of M such that $M = S \oplus T$. We claim that T satisfies the assumptions. If X is a submodule of T, then, since T is also a submodule of M, there exists a submodule Y of M such that $M = X \oplus Y$. Thus

$$T = T \cap M = T \cap (X \oplus Y) = X \oplus (T \cap Y),$$

as $X \subseteq T$. Thus every submodule of T has a complement in T. Since $\dim T < \dim M$ and $T \cap Y$ is a submodule of T, the inductive hypothesis implies that T is a direct sum of simple submodules. Hence M is a direct sum of simple submodules. □

1.24. Proposition. *If M is a semisimple module and N is a submodule, then N and M/N are semisimple.*

Proof. Assume that $M = S_1 + \cdots + S_k$, where each S_i is a simple submodule. If $\pi\colon M \to M/N$ is the canonical map, the techniques used in Schur's lemma imply that each restriction $\pi|_{S_i}$ is either zero or an isomorphism with the image. Since

$$M/N = \pi(M) = \sum_{i=1}^{k} (\pi|_{S_i})(S_i)$$

is a sum of finitely many simple submodules, it follows from Proposition 1.23 that M/N is a direct sum of finitely many simple modules.

We now prove that N is semisimple. By assumption, there exists a submodule T of M such that $M = N \oplus T$. The quotient M/T is semisimple by the previous paragraph, so it follows that

$$N \simeq N/\{0\} = N/(N \cap T) \simeq (N \oplus T)/T = M/T$$

is also semisimple. □

1.25. Definition. A finite-dimensional algebra A is **semisimple** if every finitely generated A-module is semisimple.

1.26. Proposition. *Let A be a finite-dimensional algebra. Then A is semisimple if and only if the regular representation of A is semisimple.*

Proof. Let us prove the non-trivial implication. Assume that the regular left module ${}_AA$ is semisimple. Let M be a finitely generated module, say $M = (m_1, \dots, m_k)$. The map

$$\bigoplus_{i=1}^{k} A \to M, \quad (a_1, \dots, a_k) \mapsto \sum_{i=1}^{k} a_i \cdot m_i,$$

is a surjective homomorphism of modules, where A is considered as a module with the regular representation. It follows that $\bigoplus_{i=1}^{k} A$ is semisimple. Thus M is semisimple, as it is isomorphic to the quotient of a semisimple module. □

1.27. Theorem. *Let A be a finite-dimensional semisimple algebra. Assume that the regular representation can be decomposed as $A = \bigoplus_{i=1}^{k} S_i$ where each S_i is a simple submodule. If S is a simple module, then $S \simeq S_i$ for some $i \in \{1, \dots, k\}$.*

Proof. Let $s \in S \setminus \{0\}$. The map $\varphi\colon A \to S$, $a \mapsto a \cdot s$, is a surjective module homomorphism. Since $\varphi \neq 0$, there exists $i \in \{1, \dots, k\}$ such that the restriction $\varphi|_{S_i}\colon S_i \to S$ is non-zero. By Schur's lemma, it follows that $\varphi|_{S_i}$ is an isomorphism. □

As a corollary, a finite-dimensional semisimple algebra admits only finitely many isomorphism classes of simple modules. When we say that the $S_1, \dots, S_k$ are the simple modules of an algebra, this means that the S_i are the representatives of isomorphism classes of all simple modules of the algebra, that is that each simple module is isomorphic to some S_i and, moreover, $S_i \not\simeq S_j$ whenever $i \neq j$.

1.28. Exercise. If A and B are algebras, M is a module over A and N is a module over B, then $M \oplus N$ is a module over $A \times B$ with

$$(a, b) \cdot (m, n) = (a \cdot m, b \cdot n).$$

A **division algebra** D is an algebra such that every non-zero element is invertible, that is for all $x \in D \setminus \{0\}$ there exists $y \in D$ such that $xy = yx = 1_D$. Modules over division algebras are very much like vector spaces. For example, every finitely generated module M over a division algebra has a basis. Moreover, every linearly independent subset of M can be extended to a basis of M.

1.29. Proposition. *Let D be a division algebra, V a non-zero finite-dimensional module over D, and $B = \operatorname{End}_D(V)$. Then V is a simple B-module and there exists $n \in \mathbb{Z}_{>0}$ such that ${}_BB \simeq nV$.*

Sketch of the proof. Let $\{v_1, \dots, v_n\}$ be a basis of V. A direct calculation shows that the map

$$\operatorname{End}_D(V) \to \bigoplus_{i=1}^{n} V = nV, \quad f \mapsto (f(v_1), \dots, f(v_n)),$$

is an injective homomorphism of $\operatorname{End}_D(V)$-modules. Since

$$\dim_D \operatorname{End}_D(V) = n^2 = \dim_D(nV),$$

it follows that the map is an isomorphism. Thus

$$\operatorname{End}_D(V) \simeq \bigoplus_{i=1}^{n} V.$$

It remains to show that V is simple. It is enough to prove that $V = \operatorname{End}_D(V) \cdot v = (v)$ for all $v \in V \setminus \{0\}$. Let $v \in V \setminus \{0\}$. If $w \in V$, then there exists $f \in \operatorname{End}_D(V)$ such that $f \cdot v = f(v) = w$. Thus $w \in (v)$ and therefore $V = (v)$. $\square$

The proposition states that if D is a division algebra, then D^n is a simple $M_n(D)$-module and that $M_n(D) \simeq n(D^n)$ as $M_n(D)$-modules.

1.30. Exercise. Let M, N, and X be A-modules. Prove that

$$\operatorname{Hom}_A(M \oplus N, X) \simeq \operatorname{Hom}_A(M, X) \times \operatorname{Hom}_A(N, X).$$

1.31. Theorem. *Let A be a finite-dimensional algebra, and let $S_1, \dots, S_k$ be pairwise non-isomorphic simple modules over A, and $n_1, \dots, n_k \in \mathbb{Z}_{\geq 0}$. If*

$$M \simeq n_1 S_1 \oplus \cdots \oplus n_k S_k,$$

then each n_j is uniquely determined.

Proof. Since each S_j is a simple module and $S_i \not\simeq S_j$ if $i \neq j$, Schur's lemma implies that $\operatorname{Hom}_A(S_i, S_j) = \{0\}$ whenever $i \neq j$. For each $j \in \{1, \dots, k\}$, routine calculations show that

$$\operatorname{Hom}_A(M, S_j) \simeq \operatorname{Hom}_A\left(\bigoplus_{i=1}^{k} n_i S_i, S_j\right) \simeq n_j \operatorname{Hom}_A(S_j, S_j).$$

Since M and S_j are finite-dimensional vector spaces, it follows that $\operatorname{Hom}_A(M, S_j)$ and $\operatorname{Hom}_A(S_j, S_j)$ are both finite-dimensional vector spaces. Moreover, the identity $\mathrm{id}\colon S_j \to S_j$ is a module homomorphism and hence $\dim \operatorname{Hom}_A(S_j, S_j) \geq 1$. Thus each n_j is uniquely determined, as

$$n_j = \frac{\dim_K \operatorname{Hom}_A(M, S_j)}{\dim_K \operatorname{Hom}_A(S_j, S_j)}. \qquad \square$$

1.32. Definition. If A is an algebra, the **opposite algebra** A^{op} is the vector space A with multiplication $A \times A \to A$, $(a, b) \mapsto a \cdot_{\mathrm{op}} b = ba$.

An algebra A is commutative if and only if $A = A^{\mathrm{op}}$.

1.33. Lemma. *If A is an algebra, then $A^{\mathrm{op}} \simeq \operatorname{End}_A(A)$ as algebras.*

Proof. Note that $\operatorname{End}_A(A) = \{\rho_a : a \in A\}$, where $\rho_a\colon A \to A$, $x \mapsto xa$. Indeed, if $f \in \operatorname{End}_A(A)$, then $f(1) = a \in A$. Moreover, $f(b) = f(b1) = bf(1) = ba$ and hence $f = \rho_a$. The map

$$A^{\mathrm{op}} \to \operatorname{End}_A(A), \quad a \mapsto \rho_a,$$

is bijective and it is an algebra homomorphism, as

$$\rho_a \rho_b(x) = \rho_a(\rho_b(x)) = \rho_a(xb) = x(ba) = \rho_{ba}(x). \qquad \square$$

1.34. Lemma. *If A is an algebra and $n \in \mathbb{Z}_{>0}$, then $M_n(A)^{\mathrm{op}} \simeq M_n(A^{\mathrm{op}})$ as algebras.*

Proof. Let $\psi\colon M_n(A)^{\mathrm{op}} \to M_n(A^{\mathrm{op}})$, $X \mapsto X^T$, where X^T is the transpose matrix of X. Since ψ is a bijective linear map, it is enough to see that ψ is a homomorphism. If $i, j \in \{1, \dots, n\}$, $a = (a_{ij})$ and $b = (b_{ij})$, then

$$\begin{aligned}(\psi(a)\psi(b))_{ij} &= \sum_{k=1}^{n} \psi(a)_{ik}\psi(b)_{kj} = \sum_{k=1}^{n} a_{ki} \cdot_{\mathrm{op}} b_{jk} \\ &= \sum_{k=1}^{n} b_{jk} a_{ki} = (ba)_{ji} = ((ba)^T)_{ij} = \psi(a \cdot_{\mathrm{op}} b)_{ij}. \qquad \square\end{aligned}$$

1.35. Lemma. *If S is a simple module and $n \in \mathbb{Z}_{>0}$, then*

$$\operatorname{End}_A(nS) \simeq M_n(\operatorname{End}_A(S))$$

as algebras.

Sketch of the proof. Let (φ_{ij}) be a matrix with entries in $\operatorname{End}_A(S)$. We define a map $nS \to nS$ as follows:

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mapsto \begin{pmatrix} \varphi_{11} & \cdots & \varphi_{1n} \\ \vdots & \ddots & \vdots \\ \varphi_{n1} & \cdots & \varphi_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \varphi_{11}(x_1) + \cdots + \varphi_{1n}(x_n) \\ \vdots \\ \varphi_{n1}(x_1) + \cdots + \varphi_{nn}(x_n) \end{pmatrix}.$$

The reader should prove that the map

$$M_n(\mathrm{End}_A(S)) \to \mathrm{End}_A(nS)$$

is an injective algebra homomorphism. It is surjective. Indeed, if $\psi \in \mathrm{End}_A(nS)$ and $i,j \in \{1,\dots,n\}$ one defines ψ_{ij} by

$$\psi\begin{pmatrix} x\\ 0\\ \vdots\\ 0\end{pmatrix} = \begin{pmatrix}\psi_{11}(x)\\ \psi_{21}(x)\\ \vdots\\ \psi_{n1}(x)\end{pmatrix}, \dots, \psi\begin{pmatrix} 0\\ 0\\ \vdots\\ x\end{pmatrix} = \begin{pmatrix}\psi_{1n}(x)\\ \psi_{2n}(x)\\ \vdots\\ \psi_{nn}(x)\end{pmatrix}.$$ □

1.36. Exercise. Prove Lemma 1.35.

§ 2. The Artin–Wedderburn theorem

2.1. Exercise. Let M, N, and X be A-modules. Prove that

$$\mathrm{Hom}_A(X, M\oplus N) \simeq \mathrm{Hom}_A(X,M)\times\mathrm{Hom}_A(X,N).$$

2.2. Theorem (Artin–Wedderburn). *Let A be a finite-dimensional semisimple algebra with k isomorphism classes of simple modules. Then*

$$A \simeq M_{n_1}(D_1)\times\cdots\times M_{n_k}(D_k)$$

for some $n_1,\dots,n_k \in \mathbb{Z}_{>0}$ and some division algebras $D_1,\dots,D_k$.

Proof. Decompose the regular representation as a sum of simple modules and gather the simples by isomorphism classes to get

$$A = \bigoplus_{i=1}^k n_i S_i,$$

where each S_i is simple and $S_i \not\simeq S_j$ whenever $i \neq j$. Schur's lemma implies that

$$\mathrm{End}_A(A) \simeq \mathrm{End}_A\left(\bigoplus_{i=1}^k n_iS_i\right) \simeq \prod_{i=1}^k \mathrm{End}_A(n_iS_i) \simeq \prod_{i=1}^k M_{n_i}(\mathrm{End}_A(S_i)),$$

where each $D_i = \mathrm{End}_A(S_i)$ is a division algebra by Schur's lemma. Thus

$$\mathrm{End}_A(A) \simeq \prod_{i=1}^k M_{n_i}(D_i).$$

Since $\mathrm{End}_A(A) \simeq A^{\mathrm{op}}$, it follows that

$$A = (A^{\mathrm{op}})^{\mathrm{op}} \simeq \prod_{i=1}^k M_{n_i}(D_i)^{\mathrm{op}} \simeq \prod_{i=1}^k M_{n_i}(D_i^{\mathrm{op}}).$$

Since each D_i is a division algebra, each D_i^{op} is also a division algebra. These are the division algebras mentioned in the claim. □

2.3. Corollary (Molien). *If A is a finite-dimensional complex semisimple algebra with k isomorphism classes of simple modules, then*

$$A \simeq \prod_{i=1}^{k} M_{n_i}(\mathbb{C})$$

for some $n_1, \dots, n_k \in \mathbb{Z}_{>0}$.

Proof. By the Artin–Wedderburn theorem,

$$A \simeq \prod_{i=1}^{k} M_{n_i}(\operatorname{End}_A(S_i)^{\mathrm{op}}),$$

where $S_1, \dots, S_k$ are representatives of the isomorphism classes of simple modules and each $\operatorname{End}_A(S_i)$ is a division algebra. We claim that

$$\operatorname{End}_A(S_i) = \{\lambda \operatorname{id} : \lambda \in \mathbb{C}\} \simeq \mathbb{C}$$

for all $i \in \{1, \dots, k\}$. If $f \in \operatorname{End}_A(S_i)$, then f has an eigenvalue $\lambda \in \mathbb{C}$. Since $f - \lambda \operatorname{id}$ is not invertible (and hence not an isomorphism), Schur's lemma implies that $f - \lambda \operatorname{id} = 0$, that is $f = \lambda \operatorname{id}$. Thus $\operatorname{End}_A(S_i) \to \mathbb{C}$, $f \mapsto \lambda$, is an algebra isomorphism. In particular,

$$A \simeq \prod_{i=1}^{k} M_{n_i}(\mathbb{C}). \qquad \square$$

We conclude this section with a nice application of semisimplicity.

2.4. Exercise. Let $m, n \geq 1$. There exists a unital homomorphism $M_n(\mathbb{R}) \to M_m(\mathbb{R})$ of $\mathbb{R}$-algebras if and only if n divides m.

§ 3. Simple algebras

3.1. Definition. An algebra A is said to be **simple** if $A \neq \{0\}$ and $\{0\}$ and A are the only ideals of A.

3.2. Proposition. *Let A be a finite-dimensional simple algebra. There exists a non-zero left ideal I of minimal dimension. This ideal is a simple A-module, and every simple A-module is isomorphic to I.*

Proof. Since A is finite-dimensional and A is a left ideal of A, there exists a non-zero left ideal of minimal dimension. The minimality of $\dim I$ implies that I is a simple A-module.

Let M be a simple A-module. In particular, $M \neq \{0\}$. Since

$$\operatorname{Ann}_A(M) = \{a \in A : a \cdot M = \{0\}\}$$

is an ideal of A and $1 \in A \setminus \operatorname{Ann}_A(M)$, the simplicity of A implies that $\operatorname{Ann}_A(M) = \{0\}$ and hence $I \cdot M \neq \{0\}$ (because $I \cdot m = 0$ for all $m \in M$ yields $I \subseteq \operatorname{Ann}_A(M)$ and I is non-zero, a contradiction). Let $m \in M$ be such that $I \cdot m \neq \{0\}$. The map

$$\varphi \colon I \to M, \quad x \mapsto x \cdot m,$$

is a module homomorphism. Since $I \cdot m \neq \{0\}$, the map φ is non-zero. Since both I and M are simple, Schur's lemma implies that φ is an isomorphism. $\square$

If D is a finite-dimensional division algebra, then $M_n(D)$ is a simple algebra. The previous proposition implies that the algebra $M_n(D)$ has a unique isomorphism class of simple modules. Each simple module is isomorphic to D^n.

3.3. PROPOSITION. *Let A be a finite-dimensional algebra. If A is simple, then A is semisimple.*

PROOF. Let S be the sum of the simple submodules appearing in the regular representation of A. We claim that S is an ideal of A. We know that S is a left ideal, as the submodules of the regular representation are exactly the left ideals of A. To show that $Sa \subseteq S$ for all $a \in A$ we need to prove that $Ta \subseteq S$ for every simple submodule T of A and $a \in A$. If $T \subseteq A$ is a simple submodule and $a \in A$, let $f\colon T \to Ta$, $t \mapsto ta$. Since f is a surjective module homomorphism and T is simple, it follows that either $\ker f = \{0\}$ or $\ker f = T$. If $\ker f = T$, then $f(T) = Ta = \{0\} \subseteq S$. If $\ker f = \{0\}$, then $T \simeq f(T) = Ta$ and hence Ta is simple. Hence $Ta \subseteq S$.

Since S is an ideal of A and A is a simple algebra, it follows either $S = \{0\}$ or $S = A$. Since $S \neq \{0\}$, because there exists a non-zero left ideal I of A of minimal dimension, it follows that $S = A$. Since A is finite-dimensional, it is a finite sum of simple submodules. Thus A is semisimple by Proposition 1.23. □

§ 4. Wedderburn's theorem

4.1. THEOREM (Wedderburn). *Let A be a finite-dimensional algebra. If A is simple, then $A \simeq M_n(D)$ for some $n \in \mathbb{Z}_{>0}$ and a division algebra D.*

PROOF. Since A is simple, it follows that A is semisimple. The Artin–Wedderburn theorem implies that $A \simeq \prod_{i=1}^{k} M_{n_i}(D_i)$ for some $n_1, \dots, n_k \geq 1$ and division algebras $D_1, \dots, D_k$. Moreover, A has k isomorphism classes of simple modules. Since A is simple, A has only one isomorphism class of simple modules (Proposition 3.2). Thus $k = 1$ and hence $A \simeq M_n(D)$ for some $n \in \mathbb{Z}_{>0}$ and some division algebra D. □

§ 5. Group algebras

Let K be a field, and G be a group. The **group algebra** $K[G]$ is the vector space (over K) with basis $\{g : g \in G\}$ and the algebra structure is given by the multiplication

$$\left(\sum_{g\in G} \lambda_g g\right)\left(\sum_{h\in G} \mu_h h\right) = \sum_{g,h\in G} \lambda_g \mu_h (gh).$$

Every element of $K[G]$ is a finite sum of the form $\sum_{g\in G} \lambda_g g$.

5.1. EXERCISE. If G is non-trivial, then $K[G]$ is not simple.

5.2. EXERCISE. For $n \geq 2$ let $G = C_n$ be the (multiplicative) cyclic group of order n. Prove that $K[G] \simeq K[X]/(X^n - 1)$.

5.3. EXERCISE. Let G be a finitely generated torsion-free abelian group. Prove that $K[G]$ is a domain.

5.4. EXERCISE. Let G be a group and $\alpha = \sum_{g \in G} \lambda_g g \in K[G]$. The **support** of α is the set

$$\operatorname{supp} \alpha = \{g \in G : \lambda_g \neq 0\}.$$

Prove that if $g \in G$, then $\operatorname{supp}(g\alpha) = g(\operatorname{supp} \alpha)$ and $\operatorname{supp}(\alpha g) = (\operatorname{supp} \alpha)g$.

5.5. EXERCISE. Let G be a group and H be a subgroup of G. Let $\alpha \in K[H]$. Prove that α is invertible (resp. a left zero divisor) in $K[H]$ if and only if α is invertible (resp. a left zero divisor) in $K[G]$.

5.6. EXERCISE. Let $G = C_2 = \langle g \rangle \simeq \mathbb{Z}/2$, the (multiplicative) cyclic group of order two. Note that every element of $K[G]$ is of the form $a + bg$ for some $a, b \in K$. Prove the following statements:

1) If the characteristic of K is different from two, then

$$K[G] \to K \times K, \quad a1_{K[G]} + bg \mapsto (a+b, a-b),$$

is an algebra isomorphism.

2) If the characteristic of K is two, then

$$K[G] \to \begin{pmatrix} K & K \\ 0 & K \end{pmatrix}, \quad a1 + bg \mapsto \begin{pmatrix} a+b & b \\ 0 & a+b \end{pmatrix},$$

is an algebra isomorphism onto its image.

If A is an algebra over K and $\rho \colon G \to \mathcal{U}(A)$ is a group homomorphism, where $\mathcal{U}(A)$ is the group of units of A, then the map

$$K[G] \to A, \quad \sum_{g \in G} \lambda_g g \mapsto \sum_{g \in G} \lambda_g \rho(g),$$

is an algebra homomorphism.

5.7. EXERCISE. Let $G = C_3$ be the (multiplicative) cyclic group of order three. Prove that $\mathbb{R}[G] \simeq \mathbb{R} \times \mathbb{C}$.

5.8. EXERCISE. Let $G = \langle r, s : r^3 = s^2 = 1,\ srs = r^{-1} \rangle$ be the dihedral group of six elements. Prove the following statements:

1) $\mathbb{C}[G] \simeq \mathbb{C} \times \mathbb{C} \times M_2(\mathbb{C})$.

2) $\mathbb{Q}[G] \simeq \mathbb{Q} \times \mathbb{Q} \times M_2(\mathbb{Q})$.

Maschke's theorem states that, if G is a finite group, then the group algebra $\mathbb{C}[G]$ is semisimple. By Molien's theorem,

$$\mathbb{C}[G] \simeq \prod_{i=1}^{k} M_{n_i}(\mathbb{C}),$$

where k is the number of (isomorphism classes of) simple $\mathbb{C}[G]$-modules. Moreover,

$$|G| = \dim \mathbb{C}[G] = \sum_{i=1}^{k} n_i^2.$$

5.9. Theorem. *Let G be a finite group. The number of simple modules of $\mathbb{C}[G]$ coincides with the number of conjugacy classes of G.*

Proof. By Molien's theorem, $\mathbb{C}[G] \simeq \prod_{i=1}^{k} M_{n_i}(\mathbb{C})$. Thus

$$Z(\mathbb{C}[G]) \simeq \prod_{i=1}^{k} Z(M_{n_i}(\mathbb{C})) \simeq \mathbb{C}^k.$$

In particular, $\dim Z(\mathbb{C}[G]) = k$. If $\alpha = \sum_{g\in G} \lambda_g g \in Z(\mathbb{C}[G])$, then $h^{-1}\alpha h = \alpha$ for all $h \in G$. Thus

$$\sum_{g\in G} \lambda_{hgh^{-1}} g = \sum_{g\in G} \lambda_g h^{-1} g h = \sum_{g\in G} \lambda_g g$$

and hence $\lambda_g = \lambda_{hgh^{-1}}$ for all $g, h \in G$. Note that coefficients constant on conjugacy classes give a central element. Thus a basis for $Z(\mathbb{C}[G])$ is given by elements of the form

$$\sum_{g\in K} g,$$

where K is a conjugacy class of G. Therefore $\dim Z(\mathbb{C}[G])$ is equal to the number of conjugacy classes of G. □

§ 6. Which algebras are group algebras?

6.1. Example. Let $G = C_4$ be the cyclic group of order four. Then G has four simple modules and $\mathbb{C}[G] \simeq \mathbb{C}^4$.

6.2. Example. Let $G = \mathbb{S}_3$. Then G has three simple modules and

$$\mathbb{C}[G] \simeq \mathbb{C} \times \mathbb{C} \times M_2(\mathbb{C}).$$

6.3. Open problem (Brauer). Which complex algebras are group algebras of finite groups?

The problem remains only partially understood. Examples 6.1 and 6.2 show that $\mathbb{C}^4$ and $\mathbb{C}^2 \times M_2(\mathbb{C})$ are complex group algebras.

6.4. Exercise. Is $\mathbb{C}^2 \times M_2(\mathbb{C}) \times M_3(\mathbb{C})$ a complex group algebra?

§ 7. The isomorphism problem for group algebras

Recall that if R is a unitary commutative ring and G is a group, then one defines the group ring $R[G]$. Note that $R[G]$ is a left R-module with

$$\mu\left(\sum_{g\in G}\lambda_g g\right)=\sum_{g\in G}(\mu\lambda_g)g.$$

In this section, we will briefly discuss the following natural problem:

7.1. Question (The isomorphism problem). Let R be a unitary commutative ring and let G and H be groups. Assume that $R[G]\simeq R[H]$ as unital R-algebras. Does $G\simeq H$?

For general information on Question 7.1 we refer to the survey paper [**16**].

7.2. Exercise. Prove that if G and H are isomorphic groups, then $K[G]\simeq K[H]$.

7.3. Exercise. Let G and H be groups. Prove that if $\mathbb{Z}[G]\simeq\mathbb{Z}[H]$, then $R[G]\simeq R[H]$ for any commutative ring R.

The previous exercise suggests the importance of the following instance of Question 7.1:

7.4. Question. Let G and H be groups. Does $\mathbb{Z}[G]\simeq\mathbb{Z}[H]$ imply $G\simeq H$?

For finite groups, the isomorphism problem has an affirmative answer in several important cases, including abelian, metabelian, and nilpotent groups. However, it is false in general. In 2001 Hertweck found a counterexample of order $2^{21}97^{28}$, see [**11**].

7.5. Question (The modular isomorphism problem). Let p be a prime number. Let G and H be finite p-groups and let K be a field of characteristic p. Does $K[G]\simeq K[H]$ imply $G\simeq H$?

Question 7.5 has an affirmative answer in several cases. However, this is not true in general. This question was answered by García-Lucas, Margolis and del Río [**6**]. They found two non-isomorphic groups G and H both of order 512 such that $K[G]\simeq K[H]$ for every field K of characteristic two.

§ 8. Primitive rings

We will consider (possibly non-unitary) rings. Thus a **ring** is an abelian group R with an associative multiplication $(x,y)\mapsto xy$ such that $(x+y)z=xz+yz$ and $x(y+z)=xy+xz$ for all $x,y,z\in R$. If there is an element $1\in R$ such that $x1=1x=x$ for all $x\in R$, we say that R is a **unitary ring**. A **subring** S of R is an additive subgroup of R closed under multiplication.

8.1. Example. $\mathbb{Z}$ is a (unitary) ring and $2\mathbb{Z}=\{2m:m\in\mathbb{Z}\}$ is a (non-unitary) ring.

A **left ideal** (resp. **right ideal**) is a subring I of R such that $rI \subseteq I$ (resp. $Ir \subseteq I$) for all $r \in R$. An **ideal** (also two-sided ideal) of R is a subring I of R that is both a left and a right ideal of R.

8.2. Example. If I and J are both ideals of a ring R, then the sum

$$I + J = \{x + y : x \in I, y \in J\}$$

and the intersection $I \cap J$ are both ideals of R. The product IJ, defined as the additive subgroup of R generated by $\{xy : x \in I, y \in J\}$, is also an ideal of R.

In the nonunital setting, $R^2 = RR$ normally means the additive subgroup generated by all products xy for $x, y \in R$.

8.3. Example. If R is a ring, the set $Ra = \{xa : x \in R\}$ is a left ideal of R. Similarly, the set $aR = \{ax : x \in R\}$ is a right ideal of R. The set RaR, which is defined as the additive subgroup of R generated by $\{xay : x, y \in R\}$, is a ideal of R.

8.4. Example. If R is a unitary ring, then Ra is the left ideal generated by a, aR is the right ideal generated by a and RaR is the ideal generated by a. If R is not unitary, the left ideal generated by a is $Ra + \mathbb{Z}a$, the right ideal generated by a is $aR + \mathbb{Z}a$ and the ideal generated by a is $RaR + Ra + aR + \mathbb{Z}a$.

The following exercise asks to prove the **Chinese Remainder Theorem** for arbitrary rings.

8.5. Bonus exercise. Let R be a ring and $I_1, \dots, I_n$ be ideals such that $I_j + I_k = R$ whenever $j \neq k$ and $R = I_j + R^2$ for all j. Prove that

$$R/(I_1 \cap \cdots \cap I_n) \simeq R/I_1 \times \cdots \times R/I_n.$$

In the previous exercise, the condition $R = I_j + R^2$ trivially holds in the case of rings with identity.

8.6. Definition. A ring R is said to be **simple** if $R^2 \neq \{0\}$ and the only ideals of R are $\{0\}$ and R.

The condition $R^2 \neq \{0\}$ is trivially satisfied in the case of non-zero rings with identity, as

$$1 \in R^2 = \{r_1 r_2 : r_1, r_2 \in R\}.$$

8.7. Example. Division rings are simple.

Let S be a unitary ring. Recall that for $n \geq 1$, $M_n(S)$ is the ring of $n \times n$ square matrices with entries in S. If $A = (a_{ij}) \in M_n(S)$ and E_{ij} is the matrix such that $(E_{ij})_{kl} = \delta_{ik}\delta_{jl}$, then

$$E_{ij} A E_{kl} = a_{jk} E_{il} \tag{8.1}$$

for all $i, j, k, l \in \{1, \dots, n\}$.

8.8. Example. If D is a division ring, then $M_n(D)$ is simple.

Let R be a ring. A left R-module (or module, for short) is an abelian group M together with a map $R \times M \to M$, $(r, m) \mapsto r \cdot m$, such that

$$(r + s) \cdot m = r \cdot m + s \cdot m, \qquad r \cdot (m + n) = r \cdot m + r \cdot n, \qquad r \cdot (s \cdot m) = (rs) \cdot m$$

for all $r, s \in R$, $m, n \in M$. If R has an identity 1 and $1 \cdot m = m$ holds for all $m \in M$, the module M is said to be **unitary**. If M is a unitary module, then $M = R \cdot M$.

8.9. Exercise. Let R be a simple unitary ring.

1) Prove that the center $Z(R)$ of R is a field.

2) Prove that R is an algebra over $Z(R)$.

8.10. Definition. A module M is said to be **simple** if $R \cdot M \neq \{0\}$ and the only submodules of M are $\{0\}$ and M. If M is a simple module, then $M \neq \{0\}$.

If R is a unitary ring and M is a simple module, then M is unitary.

8.11. Lemma. *Let M be a non-zero module. Then M is simple if and only if $M = R \cdot m$ for all $0 \neq m \in M$.*

Proof. Assume that M is simple. Let $m \neq 0$. Since $R \cdot m$ is a submodule of the simple module M, either $R \cdot m = \{0\}$ or $R \cdot m = M$. Let $N = \{n \in M : R \cdot n = \{0\}\}$. Since N is a submodule of M and $R \cdot M \neq \{0\}$, $N = \{0\}$. Therefore $R \cdot m = M$, as $m \neq 0$. Now assume that $M = R \cdot m$ for all $m \neq 0$. Let L be a non-zero submodule of M and let $0 \neq x \in L$. Then $M = L$, as $M = R \cdot x \subseteq L$. $\square$

8.12. Example. Let D be a division ring and let V be a non-zero right vector space over D. If $R = \mathrm{End}_D(V)$, then V is a simple R-module with $fv = f(v)$ for $f \in R$ and $v \in V$.

8.13. Example. Let $n \geq 2$. If D is a division ring and $R = M_n(D)$, then each

$$I_k = \{(a_{ij}) \in R : a_{ij} = 0 \text{ for } j \neq k\}$$

is an R-module isomorphic to D^n. Thus $M_n(D)$ is a simple ring that is not a simple $M_n(D)$-module.

8.14. Definition. A left ideal L of a ring R is said to be **minimal** if $L \neq \{0\}$ and L does not properly contain non-zero left ideals of R.

Similarly one defines minimal right ideals and minimal ideals.

8.15. Example. Let $n \geq 1$, D a division ring and $R = M_n(D)$. Then $L = RE_{11}$ is a minimal left ideal.

8.16. Example. Let L be a non-zero left ideal. If $RL \neq \{0\}$, then L is minimal if and only if L is a simple R-module.

8.17. Definition. A left (resp. right) ideal L of R is said to be **regular** if there exists $e \in R$ such that $r - re \in L$ (resp. $r - er \in L$) for all $r \in R$.

If R is a ring with identity, every left (or right) ideal is regular.

8.18. Definition. A left (resp. right) ideal I of R is said to be **maximal** if $I \neq R$ and I is not properly contained in a proper left (resp. right) ideal of R.

Similarly, one defines maximal ideals.

A standard application of Zorn's lemma proves that every non-zero unitary ring contains a maximal left ideal and a maximal right ideal.

8.19. Proposition. *Let R be a ring and M be a module. Then M is simple if and only if $M \simeq R/I$ for some maximal regular left ideal I.*

PROOF. Assume that M is simple. Then $M = R \cdot m$ for some $m \neq 0$ by Lemma 8.11. The map $\phi\colon R \to M$, $r \mapsto r \cdot m$, is a surjective homomorphism of R-modules, so the first isomorphism theorem implies that $M \simeq R/\ker\phi$.

We claim that $I = \ker\phi$ is a maximal left ideal. The correspondence theorem and the simplicity of M imply that I is a maximal left ideal (because each left ideal J such that $I \subseteq J$ yields a submodule of R/I).

We claim that I is regular. Since $M = R \cdot m$, there exists $e \in R$ such that $m = e \cdot m$. If $r \in R$, then $r - re \in I$ since $\phi(r - re) = \phi(r) - \phi(re) = r \cdot m - r \cdot (e \cdot m) = 0$.

Now assume that I is a maximal left ideal that is regular. The correspondence theorem implies that R/I has no non-zero proper submodules.

We claim that $R \cdot (R/I) \neq \{0_{R/I}\}$. Assume that $R \cdot (R/I) = \{0_{R/I}\}$ and let $r \in R$. The regularity of I implies that there exists $e \in R$ such that $r - re \in I$. Hence $r \in I$, as

$$0_{R/I} = r \cdot (e + I) = re + I = r + I,$$

a contradiction to the maximality of I. $\square$

Let R be a ring and M be a left R-module. For a subset $N \subseteq M$ we define the **annihilator** of N as the subset

$$\mathrm{Ann}_R(N) = \{r \in R : r \cdot n = 0 \text{ for all } n \in N\}.$$

8.20. EXAMPLE. $\mathrm{Ann}_{\mathbb{Z}}(\mathbb{Z}/n) = n\mathbb{Z}$.

8.21. EXERCISE. Let R be a ring and M be a module. If $N \subseteq M$ is a subset, then $\mathrm{Ann}_R(N)$ is a left ideal of R. If $N \subseteq M$ is a submodule of M, then $\mathrm{Ann}_R(N)$ is an ideal of R.

8.22. DEFINITION. A module M is said to be **faithful** if $\mathrm{Ann}_R(M) = \{0\}$.

8.23. EXAMPLE. If K is a field, then K^n is a faithful unitary $M_n(K)$-module.

8.24. EXAMPLE. If V is a vector space over a field K, then V is a faithful unitary $\mathrm{End}_K(V)$-module.

8.25. DEFINITION. A ring R is said to be **primitive** if there exists a faithful simple R-module.

Since we work with left modules, “primitive ideal” means “left primitive ideal”. Right primitive ideals are defined similarly using simple right modules.

8.26. EXERCISE. If R is a simple unitary ring, then R is primitive.

8.27. EXERCISE. If R is a commutative ring (maybe without identity), then R is primitive if and only if R is a field.

8.28. EXAMPLE. The ring $\mathbb{Z}$ is not primitive.

8.29. DEFINITION. An ideal P of a ring R is said to be **primitive** if $P = \mathrm{Ann}_R(M)$ for some simple R-module M.

As we work with left modules, our definition of primitive rings refers to **left primitive rings**. Of course, one can also define right primitive rings. As Bergman showed, there exist rings that are right primitive but not left primitive; see [**2, 3**].

8.30. Exercise. Let R be a ring and P be an ideal of R. Then P is primitive if and only if R/P is a primitive ring.

8.31. Example. Let $n \geq 1$ and $R_1, \dots, R_n$ be primitive rings and $R = R_1 \times \cdots \times R_n$. Then each
$$P_i = R_1 \times \cdots \times R_{i-1} \times \{0\} \times R_{i+1} \times \cdots \times R_n$$
is a primitive ideal of R since $R/P_i \simeq R_i$.

8.32. Lemma. *Let R be a ring. If P is a primitive ideal, there exists a regular maximal left ideal I such that $P = \{x \in R : xR \subseteq I\}$. Conversely, if I is a regular maximal left ideal, then $\{x \in R : xR \subseteq I\}$ is a primitive ideal.*

Proof. Assume that $P = \mathrm{Ann}_R(M)$ for some simple R-module M. By Proposition 8.19, there exists a regular maximal left ideal I such that $M \simeq R/I$. Then
$$P = \mathrm{Ann}_R(R/I) = \{x \in R : xR \subseteq I\}.$$

Conversely, let I be a regular maximal left ideal. By Proposition 8.19, R/I is a simple R-module. Then
$$\mathrm{Ann}_R(R/I) = \{x \in R : xR \subseteq I\}$$
is a primitive ideal. $\square$

8.33. Exercise. Maximal ideals of unitary rings are primitive.

8.34. Exercise. Prove that every primitive ideal of a commutative ring is maximal.

8.35. Bonus exercise. Let $n \geq 1$. Prove that $M_n(R)$ is primitive if and only if R is primitive.

§ 9. Jacobson's radical

9.1. Definition. Let R be a ring. The **Jacobson radical** $J(R)$ is the intersection of all the annihilators of simple left R-modules. If R does not have simple left R-modules, then $J(R) = R$.

From the definition, it follows that $J(R)$ is an ideal. Moreover,
$$J(R) = \bigcap \{P : P \text{ left primitive ideal}\}.$$

If I is an ideal of R and $n \in \mathbb{Z}_{>0}$, I^n is the additive subgroup of R generated by the set $\{y_1 \dots y_n : y_j \in I\}$.

9.2. Definition. An ideal I of R is **nilpotent** if $I^n = \{0\}$ for some $n \in \mathbb{Z}_{>0}$.

Similarly, one defines right or left nilpotent ideals. Note that an ideal I is nilpotent if and only if there exists $n \in \mathbb{Z}_{>0}$ such that $x_1 x_2 \cdots x_n = 0$ for all $x_1, \dots, x_n \in I$.

9.3. Definition. An element x of a ring is said to be **nil** (or nilpotent) if $x^n = 0$ for some $n \in \mathbb{Z}_{>0}$.

9.4. Definition. An ideal I of a ring is said to be **nil** if every element of I is nil.

Similarly, one defines right or left nil ideals. Note that every nilpotent ideal is nil, as $I^n = \{0\}$ implies $x^n = 0$ for all $x \in I$.

9.5. Example. Let

$$R = \mathbb{C}[X_1, X_2, \dots]/(X_1, X_2^2, X_3^3, \dots).$$

The ideal

$$I = (X_1, X_2, X_3, \dots)$$

is nil in R, as it is generated by nilpotent elements. However, it is not nilpotent. Indeed, if I is nilpotent, then there exists $k \in \mathbb{Z}_{>0}$ such that $I^k = \{0\}$ and hence $x_i^k = 0$ for all i, a contradiction since $x_{k+1}^k \neq 0$.

9.6. Proposition. *Let R be a ring. Then every nil left ideal is contained in $J(R)$.*

Proof. Assume that there is a nil left ideal I such that $I \not\subseteq J(R)$. There exists a simple R-module M such that $n = x \cdot m \neq 0$ for some $x \in I$ and $m \in M$. Since M is simple, $R \cdot n = M$ and hence there exists $r \in R$ such that $r \cdot n = m$. Since

$$(rx) \cdot m = r \cdot (x \cdot m) = r \cdot n = m,$$

it follows that $(rx)^k \cdot m = m$ for all $k \geq 1$, a contradiction since $rx \in I$ is a nilpotent element. $\square$

Similarly, one proves that every nil right ideal is contained in the Jacobson radical.

9.7. Definition. Let R be a ring. An element $a \in R$ is said to be **left quasi-regular** if there exists $r \in R$ such that $r + a + ra = 0$. Similarly, a is said to be **right quasi-regular** if there exists $r \in R$ such that $a + r + ar = 0$.

Let R be a ring. A direct calculation shows that the **Jacobson circle operation**

$$R \times R \to R, \quad (r, s) \mapsto r \circ s = r + s + rs,$$

is an associative operation with neutral element 0.

9.8. Example. Let $R = \mathbb{Z}/3 = \{0, 1, 2\}$ be the ring of integers modulo 3. The Jacobson circle operation of R is shown in Table 1.

Table 1. The table of a radical ring over $\mathbb{Z}/3$.

$\circ$	0	1	2
0	0	1	2
1	1	0	2
2	2	2	2

If R is unitary, an element $x \in R$ is left quasi-regular (resp. right quasi-regular) if and only if $1 + x$ is left invertible (resp. right invertible). In fact, if $r \in R$ is such that $r + x + rx = 0$, then $(1 + r)(1 + x) = 1 + r + x + rx = 1$. Conversely, if there exists $y \in R$ such that $y(1 + x) = 1$, then

$$(y - 1) \circ x = y - 1 + x + (y - 1)x = 0.$$

9.9. Example. If $x \in R$ is a nilpotent element, then $y = \sum_{n\geq 1} x^n \in R$ is left quasi-regular. In fact, if there exists N such that $x^N = 0$, then the sum defining y is finite and $y + (-x) + y(-x) = 0$. Is it also right quasi-regular?

9.10. Definition. A left ideal I of R is said to be **left quasi-regular** (resp. right quasi-regular) if every element of I is left quasi-regular (resp. right quasi-regular). A left ideal is said to be **quasi-regular** if is left and right quasi-regular.

Similarly one defines right quasi-regular ideals and quasi-regular ideals.

9.11. Lemma. *Let I be a left ideal of R. If I is left quasi-regular, then I is quasi-regular.*

Proof. Let $x \in I$. Let us prove that x is right quasi-regular. Since I is left quasi-regular, there exists $r \in R$ such that $r \circ x = r + x + rx = 0$. Since $r = -x - rx \in I$, there exists $s \in R$ such that $s \circ r = s + r + sr = 0$. Then s is right quasi-regular and

$$x = 0 \circ x = (s \circ r) \circ x = s \circ (r \circ x) = s \circ 0 = s. \qquad \square$$

For unitary rings, it is known that every element which is not a unit is contained in a maximal ideal. The proof of this fact relies on Zorn's lemma. More generally, for arbitrary rings, the following result holds.

9.12. Lemma. *Let R be a ring, and $x \in R$ be an element that is not left quasi-regular Then there exists a maximal left ideal M such that $x \notin M$. Moreover, R/M is a simple R-module and $x \notin \mathrm{Ann}_R(R/M)$.*

Proof. Let $T = \{r + rx : r \in R\}$. A straightforward calculation shows that T is a left ideal of R such that $x \notin T$ (if $x \in T$, then $r + rx = -x$ for some $r \in R$, a contradiction since x is not left quasi-regular).

The only left ideal of R containing $T \cup \{x\}$ is R. Indeed, if there exists a left ideal U containing T, then $x \notin U$, since otherwise every $r \in R$ could be written as

$$r = (r + rx) + r(-x) \in U.$$

Let $\mathcal{S}$ be the set of proper left ideals of R containing T partially ordered by inclusion. Since $T \in \mathcal{S}$, $\mathcal{S} \neq \emptyset$. If $\{K_i : i \in I\}$ is a chain in $\mathcal{S}$, then $K = \cup_{i\in I} K_i$ is an upper bound for the chain (K is a proper, as $x \notin K$). Then Zorn's lemma implies that $\mathcal{S}$ admits a maximal element M. Thus M is a maximal left ideal such that $x \notin M$.

Moreover, M is regular since $r - r(-x) \in T \subseteq M$ for all $r \in R$. Therefore R/M is a simple R-module by Proposition 8.19. Since $x \cdot (x + M) \neq M$ (if $x^2 \in M$, then $x \in M$, as $x + x^2 \in T \subseteq M$), it follows that $x \notin \mathrm{Ann}_R(R/M)$. $\square$

9.13. Exercise. Let R be a ring and $x \in R$ be an element that is not left quasi-regular. Prove that $x \notin J(R)$.

9.14. Theorem. *Let R be a ring and $x \in R$. The following statements are equivalent:*

1) *The left ideal generated by x is quasi-regular.*
2) *Rx is quasi-regular.*
3) *$x \in J(R)$.*

Proof. The implication (1) $\implies$ (2) is trivial, as Rx is included in the left ideal generated by x.

We now prove (2) $\implies$ (3). If $x \notin J(R)$, by definition, there exists a simple R-module M such that $x \cdot m \neq 0$ for some $m \in M$. The simplicity of M implies that $(Rx) \cdot m = M$. Thus there exists $r \in R$ such that $(rx) \cdot m = -m$. There is an element $s \in R$ such that $s + rx + s(rx) = 0$ and hence

$$-m = (rx) \cdot m = (-s - srx) \cdot m = -s \cdot m + s \cdot m = 0,$$

a contradiction.

Finally, we prove (3) $\implies$ (1). Let $x \in J(R)$ and I be the left ideal of R generated by x. Since $I \subseteq J(R)$ and every element of $J(R)$ is left quasi-regular (see Exercise 9.13), I is left quasi-regular. By Lemma 9.11, I is quasi-regular. □

The theorem implies the following corollary.

9.15. Corollary. *If R is a ring, then $J(R)$ is a quasi-regular ideal that contains every quasi-regular left ideal.*

Proof. By Exercise 9.13, $J(R)$ is left quasi-regular. By Lemma 9.11, $J(R)$ is quasi-regular. Let I be a left ideal of R that is quasi-regular. Let $x \in I$ and J be the left ideal of R generated by x. Since $J \subseteq I$, J is quasi-regular. By Theorem 9.14, $x \in J(R)$. □

The following exercise uses Zorn's lemma and will be used in the proof of Theorem 9.17.

9.16. Exercise. Let R be a ring. Prove that every proper left ideal of R that is regular is contained in a maximal left ideal that is regular.

9.17. Theorem. *Let R be a ring such that $J(R) \neq R$. Then*

$$J(R) = \bigcap \{I : I \textit{ regular maximal left ideal of } R\}.$$

Proof. Let

$$K = \bigcap \{I : I \text{ regular maximal left ideal of } R\}.$$

Let us prove that $K \subseteq J(R)$. By Lemma 9.11, it is enough to prove that K is left quasi-regular. Let $a \in K$ and $T = \{r + ra : r \in R\}$. If $T = R$, then $-a = r + ra$ for some $r \in R$ and hence a is left quasi-regular. So we need to prove that $T = R$. Note that T is a regular left ideal with $e = -a$ (see Definition 8.17). If $T \neq R$, then T is contained in a maximal left ideal J by the previous exercise. Then $a \in K \subseteq J$ and hence $ra \in J$ for all $r \in R$. Since $r + ra \in T \subseteq J$ for all $r \in R$, it follows that $J = R$, a contradiction. Therefore $T = R$.

Now we prove that $J(R) \subseteq K$. By Proposition 8.19,

$$J(R) = \bigcap \{\mathrm{Ann}_R(R/I) : I \text{ regular maximal left ideal of } R\}.$$

Let I be a regular maximal left ideal. If $r \in J(R) \subseteq \mathrm{Ann}_R(R/I)$, then, since I is regular, there exists $e \in R$ such that $r - re \in I$. Since $r \in \mathrm{Ann}_R(R/I)$,

$$re + I = r(e + I) \subseteq I,$$

Thus $re \in I$ and hence $r \in I$. Therefore $J(R) \subseteq K$. □

9.18. Example. Each maximal ideals of $\mathbb{Z}$ is of the form $p\mathbb{Z} = \{pm : m \in \mathbb{Z}\}$ for some prime number p. Thus $J(\mathbb{Z}) = \cap_p p\mathbb{Z} = \{0\}$.

We now review some basic results useful to compute radicals.

9.19. Proposition. *Let $\{R_i : i \in I\}$ be a family of rings. Then*

$$J\left(\prod_{i\in I} R_i\right) = \prod_{i\in I} J(R_i).$$

Proof. Let $R = \prod_{i\in I} R_i$ and $x = (x_i)_{i\in I} \in R$. The left ideal Rx is quasi-regular if and only if each left ideal $R_i x_i$ is quasi-regular in R_i, as x is quasi-regular in R if and only if each x_i is quasi-regular in R_i. Thus $x \in J(R)$ if and only if $x_i \in J(R_i)$ for all $i \in I$. □

For the next result, we shall need a lemma.

9.20. Lemma. *Let R be a ring and $x \in R$. If $-x^2$ is a left quasi-regular element, then so is x.*

Proof. Let $r \in R$ be such that $r + (-x^2) + r(-x^2) = 0$ and $s = r - x - rx$. Then x is left quasi-regular, as

$$\begin{aligned} s + x + sx &= (r - x - rx) + x + (r - x - rx)x \\ &= r - x - rx + x + rx - x^2 - rx^2 = r - x^2 - rx^2 = 0. \end{aligned}$$

□

9.21. Proposition. *If I is an ideal of R, then $J(I) = I \cap J(R)$.*

Proof. Note that $I \cap J(R)$ is an ideal of I. Let $x \in I \cap J(R)$ and $r \in R$. Since rx is left quasi-regular in R, there exists $s \in R$ such that $s + rx + srx = 0$. Since $s = -rx - srx \in I$, rx is left quasi-regular in I. Thus $I \cap J(R) \subseteq J(I)$.

Let $x \in J(I) \subseteq I$ and $r \in R$. Since $-(rx)^2 = (-rxr)x \in I(J(I)) \subseteq J(I)$, the element $-(rx)^2$ is left quasi-regular in I. Thus rx is left quasi-regular by Lemma 9.20. □

§ 10. Radical rings

10.1. Definition. A ring R is said to be **radical** if $J(R) = R$.

10.2. Example. If R is a ring, then $J(R)$ is a radical ring, by Proposition 9.21.

10.3. Example. The Jacobson radical of $\mathbb{Z}/8$ is $\{0, 2, 4, 6\}$.

There are several characterizations of radical rings.

10.4. Theorem. *Let R be a ring. The following statements are equivalent:*

1) *R is radical.*
2) *R admits no simple R-modules.*
3) *R does not have regular maximal left ideals.*
4) *R does not have primitive left ideals.*
5) *Every element of R is quasi-regular.*
6) *$(R, \circ)$ is a group.*

10.5. Exercise. Prove Theorem 10.4.

10.6. Example. Let

$$A = \left\{ \frac{2x}{2y+1} : x, y \in \mathbb{Z} \right\}.$$

Then A is a radical ring, as the inverse of an arbitrary element of the form $\frac{2x}{2y+1}$ with respect to the Jacobson circle operation is

$$\left(\frac{2x}{2y+1}\right)' = \frac{-2x}{2(x+y)+1}.$$

§ 11. Henriksen's theorem

There are (non-unitary) rings with no maximal ideals. An easy example is given in the following exercise:

11.1. Exercise.

1) Prove that the additive group of rational numbers is an abelian group with no maximal subgroups.

2) Prove that the additive group $\mathbb{Q}$ of rationals with zero multiplication (that is, $xy = 0$ for all $x, y \in \mathbb{Q}$) is a ring with no maximal ideals.

Recall that the **characteristic of a ring** is defined as the least positive integer n such that $nx = 0$ for all x. If no such n exists, then we say that the ring is of **characteristic zero**.

11.2. Exercise. Let R be a non-zero ring and p be a prime number. If $px = 0$ for all $x \in R$, then R has characteristic p.

We now characterize commutative rings with no maximal ideals. We begin with some exercises.

11.3. Exercise. Let R be a commutative ring and I be an ideal of R. Prove that I is maximal if and only if R/I is a field or a ring isomorphic to $\mathbb{Z}/p$ with zero multiplication for some prime number p.

11.4. Exercise. Let R be a commutative ring. Prove that $J(R)$ equals the intersection of maximal ideals such that R/M is a field.

The following result appeared in [8].

11.5. Theorem (Henriksen). *Let R be a commutative ring. Then R has no maximal ideals if and only if $J(R) = R$ and $R^2 + pR = R$ for all prime number p.*

Proof. Assume first that R has no maximal ideals. Then $J(R) = R$ by Exercise 11.4. Let p be a prime number such that $I = R^2 + pR \neq R$. Then I is a proper ideal of R. Let $\pi\colon R \to R/I$ be the canonical map. Since $R^2 \subseteq I$, $0 = \pi(xy) = \pi(x)\pi(y)$ for all $x, y \in R$. Thus R/I has zero multiplication. Moreover, by Exercise 11.2, R/I has characteristic p, as $pR \subseteq I$. Thus R/I is a vector space over the field $\mathbb{Z}/p$. Let $\{x_\alpha : \alpha \in \Lambda\}$ be a basis of R/I. Every element $x \in R/I$ can be written uniquely as a finite sum of the form $x = \sum \lambda_\alpha x_\alpha$ for scalars λ_α. Let A be the ring with underlying additive group $\mathbb{Z}/p$ and zero multiplication. For a fixed $\beta \in \Lambda$, the map

$$\gamma\colon R/I \to A, \quad x = \sum \lambda_\alpha x_\alpha \mapsto \lambda_\beta$$

is a surjective ring homomorphism. The composition $f = \gamma\pi\colon R \to R/I \to A$ is a ring homomorphism. By Exercise 11.3, $\ker f$ is a maximal ideal, a contradiction.

Conversely, let M be a maximal ideal of R. If R/M is a field, then $J(R) \subseteq M \neq R$, a contradiction. By Exercise 11.3, there exists a prime number p such that $R/M \simeq \mathbb{Z}/p$ as abelian groups and zero multiplication (i.e. $xy \in M$ for all $x, y \in R$). Let $\pi\colon R \to R/M$ be the canonical map. Note that $R^2 \subseteq M$. Moreover, $pR \subseteq M$, as $\pi(px) = p\pi(x) = 0$ for all $x \in R$. Thus $R^2 + pR \subseteq M \neq R$, a contradiction. □

We now present a non-trivial concrete example of a ring with no maximal ideals. For that purpose, we will use the field of fractions $\mathbb{R}(X)$ of the real polynomial ring $\mathbb{R}[X]$.

11.6. Exercise. Let R be the set of rational real functions of the form $f(X)/g(X)$, where $f(X), g(X) \in \mathbb{R}(X)$ and $g(0) \neq 0$. Prove the following statements:

1) R is an integral domain with a unique maximal ideal $M = XR$.

2) M has no maximal ideals.

11.7. Definition. A ring R is said to be **nil** if for every $x \in R$ there exists $n = n(x)$ such that $x^n = 0$.

11.8. Exercise. Prove that every nil ring is radical.

11.9. Exercise. Let $\mathbb{R}[\![X]\!]$ be the ring of power series with real coefficients. Prove that the ideal $X\mathbb{R}[\![X]\!]$ consisting of power series with zero constant term is a radical ring that is not nil.

11.10. Theorem. *If R is a ring, then $J(R/J(R)) = \{0\}$.*

Proof. If R is radical, the result is trivial. Suppose then that $J(R) \neq R$. Let M be a simple R-module. Then M is a simple module over $R/J(R)$ with

$$(x + J(R)) \cdot m = x \cdot m, \quad x \in R,\, m \in M. \tag{11.1}$$

Let us prove first that (11.1) is well-defined: If $r, s \in R$ are such that $r + J(R) = s + J(R)$, then $r - s \in J(R)$. Since $J(R)$ is the intersection of the annihilators of simple R-modules (see Definition 9.1) and M is a simple R-module, $(r - s) \cdot M = \{0\}$. Thus $(r - s) \cdot m = 0$ for all $m \in M$. Hence $r \cdot m = s \cdot m$ for all $m \in M$.

We now prove that M is a simple $R/J(R)$-module. In fact, if N is a submodule of the $R/J(R)$-module M, then

$$(r + J(R)) \cdot N \subseteq N$$

for all $r \in R$. Thus $r \cdot N \subseteq N$ for all $r \in R$. Thus N is a submodule of the R-module M. Since M is simple, either $N = \{0\}$ or $N = M$.

Finally, if $r + J(R) \in J(R/J(R))$, then $r \cdot M = (r + J(R)) \cdot M = \{0\}$. Then $r \in J(R)$, as r annihilates any simple module over R. □

11.11. Theorem. *Let R be a ring and $n \in \mathbb{Z}_{>0}$. Then $J(M_n(R)) = M_n(J(R))$.*

Proof. For clarity, we will prove the result for $n = 3$. The extension to general n is straightforward and left as an exercise for the reader.

We first prove that $J(M_3(R)) \subseteq M_3(J(R))$. If $J(R) = R$, the theorem is clear. Let us assume that $J(R) \neq R$ and let $J = J(R)$. If M is a simple R-module, then M^3 is a simple $M_3(R)$-module with the usual multiplication. Let $x=(x_{ij})\in J(M_3(R))$ and $m_1,m_2,m_3\in M$. Then

$$x\begin{pmatrix}m_1\\m_2\\m_3\end{pmatrix}=0.$$

In particular, $x_{ij}\in\operatorname{Ann}_R(M)$ for all $i,j\in\{1,2,3\}$. Hence $x\in M_3(J)$.

We now prove that $M_3(J)\subseteq J(M_3(R))$. Let

$$J_2=\begin{pmatrix}0&J&0\\0&J&0\\0&J&0\end{pmatrix}\quad\text{and}\quad x=\begin{pmatrix}0&x_1&0\\0&x_2&0\\0&x_3&0\end{pmatrix}\in J_2.$$

Since x_2 is quasi-regular, there exists $y_2\in R$ such that $x_2+y_2+x_2y_2=0$. For

$$y=\begin{pmatrix}0&0&0\\0&y_2&0\\0&0&0\end{pmatrix},$$

let

$$u=x+y+xy=\begin{pmatrix}0&x_1+x_1y_2&0\\0&x_2+y_2+x_2y_2&0\\0&x_3+x_3y_2&0\end{pmatrix}=\begin{pmatrix}0&x_1+x_1y_2&0\\0&0&0\\0&x_3+x_3y_2&0\end{pmatrix}.$$

Since $u^2=0$, x is right quasi-regular, as

$$x+(y-u-yu)+x(y-u-uy)=0,$$

and therefore J_2 is right quasi-regular. Similarly one proves that both J_1 and J_3 are right quasi-regular. Thus $J_i\subseteq J(M_3(R))$ for all $i\in\{1,2,3\}$. In conclusion,

$$J_1+J_2+J_3\subseteq J(M_3(R))$$

and therefore $M_3(J)\subseteq J(M_3(R))$. □

11.12. Exercise. Let R be a ring and $n\in\mathbb{Z}_{>0}$. Prove that $J(M_n(R))=M_n(J(R))$.

11.13. Exercise. Let R be a unitary ring. Then

$$J(R)=\bigcap\{M:M\text{ is a left maximal ideal}\}.$$

11.14. Exercise. Let R be a unitary ring. The following statements are equivalent:

1) $x\in J(R)$.
2) $x\cdot M=\{0\}$ for all simple R-module M.
3) $x\in P$ for all primitive left ideal P.
4) $1+rx$ is invertible for all $r\in R$.
5) $1+\sum_{i=1}^n r_ixs_i$ is invertible for all n and all $r_i,s_i\in R$.
6) x belongs to every maximal ideal maximal.

The following exercise is entirely optional. It somewhat shows a recent application of radical rings to solutions of the celebrated Yang–Baxter equation.

11.15. Bonus exercise. A pair (X,r) is a **solution** to the Yang–Baxter equation if X is a set and $r\colon X\times X\to X\times X$ is a bijective map such that

$$(r\times \mathrm{id})\circ(\mathrm{id}\times r)\circ(r\times \mathrm{id})=(\mathrm{id}\times r)\circ(r\times \mathrm{id})\circ(\mathrm{id}\times r).$$

The solution (X,r) is said to be **involutive** if $r^2=\mathrm{id}$. By convention, we write

$$r(x,y)=(\sigma_x(y),\tau_y(x)).$$

The solution (X,r) is said to be **non-degenerate** $\sigma_x\colon X\to X$ and $\tau_x\colon X\to X$ are bijective for all $x\in X$.

1) Let X be a set and $\sigma\colon X\to X$ be a bijective map. Prove that the pair (X,r), where $r(x,y)=(\sigma(y),\sigma^{-1}(x))$, is an involutive non-degenerate solution.

Let R be a radical ring. For $x,y\in R$ let

$$\lambda_x(y)=-x+x\circ y=xy+y,$$
$$\mu_y(x)=\lambda_x(y)'\circ x\circ y=(xy+y)'x+x$$

Prove the following statements:

2) $\lambda\colon (R,\circ)\to \mathrm{Aut}(R,+)$, $x\mapsto\lambda_x$, is a group homomorphism.
3) $\mu\colon (R,\circ)\to \mathrm{Aut}(R,+)$, $y\mapsto\mu_y$, is a group antihomomorphism.
4) The map

$$r\colon R\times R\to R\times R,\quad r(x,y)=(\lambda_x(y),\mu_y(x)),$$

is an involutive non-degenerate solution to the Yang–Baxter equation.

11.16. Exercise. If D is a division ring and $R=D[X_1,\dots,X_n]$, then $J(R)=\{0\}$.

11.17. Example. A commutative and unitary ring R is **local** if it contains only one maximal ideal. If R is a local ring and M is its maximal ideal, then $J(R)=M$. Some particular cases:

1) If K is a field and $R=K[\![X]\!]$, then $J(R)=(X)$.
2) If p is a prime number and $R=\mathbb{Z}/p^n$, then $J(R)=(p)$.

We finish the discussion on the Jacobson radical with some results in the case of unitary algebras. We first need an application of Zorn's lemma.

11.18. Exercise. Let I be a proper left ideal that is left regular. Prove that I is contained in a maximal left ideal which is regular.

11.19. Proposition. *Let A be a K-algebra and I be a subset of A. Then I is a regular maximal left ideal of the algebra A if and only if I is a regular maximal left ideal of the ring A.*

Proof. Let I be a left regular maximal ideal of the ring A. We claim that $\lambda I\subseteq I$ for all $\lambda\in K$. Assume that $\lambda I\not\subseteq I$ for some λ. Then $I+\lambda I$ is an ideal of the ring A that contains I, as

$$a(I+\lambda I)=aI+a(\lambda I)\subseteq I+\lambda(aI)\subseteq I+\lambda I.$$

Since I is maximal, it follows that $I+\lambda I=A$. The left regularity of I implies that there exists $e\in A$ such that $a-ae\in I$ for all $a\in A$. Write $e=x+\lambda y$ for $x,y\in I$. Then

$$e^2=e(x+\lambda y)=ex+e(\lambda y)=ex+(\lambda e)y\in I.$$

Since $e-e^2\in I$ and $e^2\in I$, it follows that $e\in I$. Thus $A=I$, as $a-ae\in I$ for all $a\in A$, a contradiction.

Conversely, if I is a left regular maximal ideal of the algebra A, then I is a left regular ideal of the ring A. We claim that I is a maximal left ideal of the ring of A. There exists a regular maximal left ideal M of the ring A that contains I. Since M is regular, it follows that M is a regular maximal ideal of the algebra A. Thus $M=I$ because I is a maximal left ideal of the algebra A. □

For algebras, the Jacobson radical of an algebra can be defined as the intersection of the left ideals (of the algebra) that are maximal and regular. The previous proposition then implies that the Jacobson radical of an algebra coincides with the Jacobson radical of the underlying ring.

§ 12. Amitsur's theorem

We now prove an important result of Amitsur that has several interesting applications. We first need a lemma.

12.1. Lemma. *Let A be an algebra with one and let $x\in J(A)$. Then x is algebraic if and only if x is nilpotent.*

Proof. Since x is algebraic, there exist $a_0,\dots,a_n\in K$ not all zero such that

$$a_0+a_1x+\cdots+a_nx^n=0.$$

Let r be the smallest integer such that $a_r\neq 0$. Then

$$x^r(1+b_1x+\cdots+b_mx^m)=0,$$

for some $b_1,\dots,b_m\in K$. Since $1+b_1x+\cdots+b_mx^m$ is a unit by Exercise 11.14, it follows that $x^r=0$. □

An application:

12.2. Proposition. *If A is an algebraic algebra with one, then $J(A)$ is the largest nil ideal of A.*

Proof. The previous lemma implies that $J(A)$ is a nil ideal. Proposition 9.6 now implies that $J(A)$ is the largest nil ideal of A. □

12.3. Theorem (Amitsur). *Let A be a K-algebra with one such that $\dim_K A<|K|$ (as cardinals). Then $J(A)$ is the largest nil ideal of A.*

Proof. If K is finite, then A is a finite-dimensional algebra. In particular, A is algebraic and hence $J(A)$ is a nil ideal by Proposition 12.2.

Assume that K is infinite and let $a\in J(A)$. Exercise 11.14 implies that every element of the form $1-\lambda^{-1}a$, $\lambda\in K\setminus\{0\}$, is invertible. Thus

$$a-\lambda=-\lambda(1-\lambda^{-1}a)$$

is invertible for all $\lambda\in K\setminus\{0\}$. Let $S=\{(a-\lambda)^{-1}:\lambda\in K\setminus\{0\}\}$. Since

$$(a-\lambda)^{-1}=(a-\mu)^{-1}\iff\lambda=\mu,$$

it follows that $|S|=|K\setminus\{0\}|=|K|>\dim_K A$. Then S is a linearly dependent set, so there are $\beta_1,\dots,\beta_n\in K$ not all zero and distinct elements $\lambda_1,\dots,\lambda_n\in K$ such that

$$\sum_{i=1}^{n}\beta_i(a-\lambda_i)^{-1}=0. \tag{12.1}$$

Multiplying (12.1) by $\prod_{i=1}^{n}(a-\lambda_i)$ we get

$$\sum_{i=1}^{n}\beta_i\prod_{j\neq i}(a-\lambda_j)=0.$$

We claim that a is algebraic over K. Indeed,

$$f(X)=\sum_{i=1}^{n}\beta_i\prod_{j\neq i}(X-\lambda_j)$$

is non-zero, as, for example, if $\beta_1\neq 0$, then $f(\lambda_1)=\beta_1(\lambda_1-\lambda_2)\cdots(\lambda_1-\lambda_n)\neq 0$ and $f(a)=0$. Since $a\in J(A)$ is algebraic, it follows a is nilpotent by Lemma 12.1. □

Amitsur's theorem implies the following result.

12.4. Corollary. *Let K be a non-countable field. If A is an algebra over K with a countable basis, then $J(A)$ is the largest nil ideal of A.*

§ 13. Jacobson's conjecture

We now conclude the lecture with two big open problems related to the Jacobson radical. The first one is **Jacobson's conjecture**.

13.1. Open problem (Jacobson). Let R be a noetherian ring. Is then

$$\bigcap_{n\geq 1}J(R)^n=\{0\}?$$

Open problem 13.1 was originally formulated by Jacobson in 1956 [**13**] for one-sided noetherian rings. In 1965 Herstein [**9**] found a counterexample in the case of one-sided noetherian rings and reformulated the conjecture as it appears here.

13.2. Exercise (Herstein). Let D be the ring of rationals with odd denominators. Let $R=\begin{pmatrix}D&\mathbb{Q}\\0&\mathbb{Q}\end{pmatrix}$. Prove that R is right noetherian and $J(R)=\begin{pmatrix}J(D)&\mathbb{Q}\\0&0\end{pmatrix}$. Prove that $J(R)^n\supseteq\begin{pmatrix}0&\mathbb{Q}\\0&0\end{pmatrix}$ and hence $\bigcap_n J(R)^n$ is non-zero.

§ 14. Köthe's conjecture

The following problem is maybe the most important open problem in non-commutative ring theory.

14.1. Open problem (Köthe). Let R be a ring. Is the sum of two arbitrary nil left ideals of R nil?

Open problem 14.1 is the well-known **Köthe's conjecture**. The conjecture was first formulated in 1930, see [14]. It is known to be true in several cases. In full generality, the problem is still open. In [15] Krempa proved that the following statements are equivalent:

1) Köthe's conjecture is true.
2) If R is a nil ring, then $R[X]$ is a radical ring.
3) If R is a nil ring, then $M_2(R)$ is a nil ring.
4) Let $n \geq 2$. If R is a nil ring, then $M_n(R)$ is a nil ring.

In 1956 Amitsur formulated the following conjecture, see for example [1]: If R is a nil ring, then $R[X]$ is a nil ring. In [22] Smoktunowicz found a counterexample to Amitsur's conjecture. This counterexample suggests that Köthe's conjecture might be false. A simplification of Smoktunowicz' example appears in [19]. See [23, 24] for more information on Köthe's conjecture and related topics.

§ 15. Gilmer's theorem

Hilbert's theorem states that if R is a noetherian commutative unitary ring, then $R[X]$ is noetherian. Following [7], we now present the converse of Hilbert's theorem.

15.1. Theorem (Gilmer). *Let R be a commutative ring. If $R[X]$ is noetherian, then R is unitary.*

Proof. Let $a \in R$. For $m \geq 0$, let

$$\begin{aligned} I_m &= (a, aX, aX^2, \dots, aX^m) \\ &= R[X]a + R[X]aX + \cdots + R[X]aX^m + \mathbb{Z}a + \mathbb{Z}aX + \cdots + \mathbb{Z}aX^m. \end{aligned}$$

Then $I_0 \subseteq I_1 \subseteq \cdots I_m \subseteq I_{m+1} \subseteq \cdots$ is a sequence of ideals of $R[X]$. Since $R[X]$ is noetherian, $I_n = I_{n+1}$ for some n. In particular, $aX^{n+1} \in I_{n+1} = I_n$. Thus

$$aX^{n+1} = \sum_{i=1}^{n+1} aX^{i-1} f_i(X) + \sum_{i=1}^{n+1} k_i aX^{i-1}$$

for some $f_1(X), \dots, f_n(X) \in R[X]$ and $k_1, \dots, k_n \in \mathbb{Z}$. Comparing the coefficient of X^{n+1} one gets that $a = ar$ for some $r \in R$. Thus

$$\text{for every } a \in R \text{ there exists } r \in R \text{ such that } a = ra. \tag{15.1}$$

Claim. For every $a_1, \dots, a_n \in R$ there exists $r \in R$ such that $a_i = ra_i$ for all i.

We proceed by induction on n. The case $n = 1$ is (15.1). Assume that the result holds for $n - 1 \geq 1$. By the inductive hypothesis, there exists $r_1 \in R$ such that $a_i = r_1 a_i$ for all $i \in \{1, \dots, n-1\}$. Moreover, there exists $r_2 \in R$ such that $a_n = r_2 a_n$. Let $r = r_1 + r_2 - r_1 r_2$. Then

$$ra_n = r_1 a_n + r_2 a_n - r_1 r_2 a_n = r_1 a_n + a_n - r_1 a_n = a_n.$$

Moreover, for $i \in \{1, \dots, n-1\}$,

$$ra_i = r_1 a_i + r_2 a_i - r_1 r_2 a_i = a_i + r_2 a_i - r_2 r_1 a_i = a_i + r_2 a_i - r_2 a_i = a_i.$$

We now finish the proof of the theorem. Let $R[X] \to R$, $f(X) \mapsto f(0)$, be an evaluation map. Since it is a surjective ring homomorphism, R is noetherian. In particular, R is finitely generated, say

$$R = (a_1, \dots, a_n) = Ra_1 + \cdots + Ra_n + \mathbb{Z}a_1 + \cdots + \mathbb{Z}a_n$$

for some $a_1, \dots, a_n \in R$.

We now prove that the element r from the claim we proved turns R into a unitary ring, that is $r = 1_R$. We need to show that $rb = b$ for all $b \in R$. If $b \in R$, then

$$b = t_1 a_1 + \cdots + t_n a_n + m_1 a_1 + \cdots + m_n a_n$$

for some $t_1, \dots, t_n \in R$ and $m_1, \dots, m_n \in \mathbb{Z}$. Since $a_i = r a_i$ for all $i \in \{1, \dots, n\}$, it immediately follows that $rb = b$. $\square$

15.2. Example. The polynomial ring $(2\mathbb{Z})[X]$ is not noetherian, as the ring $2\mathbb{Z}$ is not unitary.

§ 16. Artinian modules

16.1. Definition. Let R be a ring. A module N is **artinian** if every decreasing sequence $N_1 \supseteq N_2 \supseteq \cdots$ of submodules of N stabilizes, that is there exists $n \in \mathbb{Z}_{>0}$ such that $N_n = N_{n+k}$ for all $k \in \mathbb{Z}_{\geq 0}$.

Let X be a set and $\mathcal{S}$ be a set of subsets of X. We say that $A \in \mathcal{S}$ is a **minimal element** of $\mathcal{S}$ if there is no $Y \in \mathcal{S}$ such that $Y \subsetneq A$.

16.2. Proposition. *A module N is artinian if and only if every non-empty subset of submodules of N contains a minimal element.*

Proof. Assume that N is artinian. Let $\mathcal{S}$ be a non-empty set of submodules of N. Suppose that $\mathcal{S}$ has no minimal element and let $N_1 \in \mathcal{S}$. Since N_1 is not minimal, there exists $N_2 \in \mathcal{S}$ such that $N_1 \supsetneq N_2$. Now assume we have the submodules

$$N_1 \supsetneq N_2 \supsetneq \cdots \supsetneq N_k.$$

Since N_k is not minimal, there exists N_{k+1} such that $N_k \supsetneq N_{k+1}$. This procedure produces a sequence $N_1 \supsetneq N_2 \supsetneq \cdots$ that cannot stabilize, a contradiction.

If $N_1 \supseteq N_2 \supseteq \cdots$ is a sequence of submodules, then $\mathcal{S} = \{N_j : j \geq 1\}$ has a minimal element, say N_n. Then $N_n = N_{n+k}$ for all k. $\square$

A module N is **noetherian** if for every sequence $N_1 \subseteq N_2 \subseteq \cdots$ of submodules of N there exists $n \in \mathbb{Z}_{>0}$ such that $N_n = N_{n+k}$ for all $k \in \mathbb{Z}_{\geq 0}$.

16.3. Exercise. Let M be a module. The following statements are equivalent:

1) M is noetherian.
2) Every submodule of M is finitely generated.
3) Every non-empty subset $\mathcal{S}$ of submodules of M contains a maximal element, that is an element $X \in \mathcal{S}$ such that there is no $Z \in \mathcal{S}$ such that $X \subsetneq Z$.

16.4. Exercise. Prove that a ring R is left noetherian if every sequence of left ideals $I_1 \subseteq I_2 \subseteq \cdots$ stabilizes.

16.5. Exercise. Let

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

be an exact sequence of modules. Prove that B is noetherian (resp. artinian) if and only if A and C are noetherian (resp. artinian).

16.6. Definition. A ring R is **left artinian** if the module ${}_RR$ is artinian.

Similarly one defines right artinian rings.

16.7. Example. The ring $\mathbb{Z}$ is noetherian. It is not artinian, as the sequence

$$2\mathbb{Z} \supseteq 4\mathbb{Z} \supseteq 8\mathbb{Z} \supseteq \cdots$$

does not stabilize.

16.8. Exercise. Prove that a ring R is left artinian if every sequence of left ideals $I_1 \supseteq I_2 \supseteq \cdots$ stabilizes.

16.9. Theorem. *If R is a left artinian ring, then $J(R)$ is nilpotent.*

Proof. Let $J = J(R)$. Since R is a left artinian ring, the sequence $(J^m)_{m\in\mathbb{Z}_{>0}}$ of left ideals stabilizes. There exists $k \in \mathbb{Z}_{>0}$ such that $J^k = J^l$ for all $l \geq k$. We claim that $J^k = \{0\}$. If $J^k \neq \{0\}$ let $\mathcal{S}$ the set of left ideals I such that $J^kI \neq \{0\}$. Since

$$J^kJ^k = J^{2k} = J^k \neq \{0\},$$

the set $\mathcal{S}$ is non-empty. Since R is left artinian, $\mathcal{S}$ has a minimal element I_0. Since $J^kI_0 \neq \{0\}$, let $x \in I_0 \setminus \{0\}$ be such that $J^kx \neq \{0\}$. Moreover, J^kx is a left ideal of R contained in I_0 and such that $J^kx \in \mathcal{S}$, as $J^k(J^kx) = J^{2k}x = J^kx \neq \{0\}$. The minimality of I_0 implies that, $J^kx = I_0$. In particular, there exists $r \in J^k \subseteq J$ such that $rx = x$. Since $-r \in J(R)$ is left quasi-regular, there exists $s \in R$ such that $s - r - sr = 0$. Thus

$$x = rx = (s - sr)x = sx - s(rx) = sx - sx = 0,$$

a contradiction. □

16.10. Corollary. *Let R be a left artinian ring. Each nil left ideal is nilpotent and $J(R)$ is the unique maximal nilpotent ideal of R.*

Proof. Let L be a nil left ideal of R. By Proposition 9.6, L is contained in $J(R)$. Thus L is nilpotent, as $J(R)$ is nilpotent by Theorem 16.9. □

§ 17. The Jordan–Hölder theorem

17.1. Definition. A **composition series** of the module M is a sequence

$$\{0\} = M_0 \subsetneq M_1 \subsetneq M_2 \subsetneq \cdots \subsetneq M_n = M$$

of submodules of M such that each M_i/M_{i-1} is non-zero and has no non-zero proper submodules. In this case n is the length of the composition series.

The previous definition makes sense also for non-unitary rings. That is why it is required that each quotient M_i/M_{i-1} has no proper submodules.

17.2. Theorem. *A non-zero module admits a composition series if and only if it is artinian and noetherian.*

Proof. Let M be a non-zero module and let $\{0\} = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$ be a composition series for M. We claim that each M_i is artinian and noetherian. We proceed by induction on i. The case $i = 0$ is trivial. Let us assume that M_i is artinian and noetherian. Since M_i/M_{i+1} has no proper submodules and the sequence

$$0 \longrightarrow M_i \longrightarrow M_{i+1} \longrightarrow M_{i+1}/M_i \longrightarrow 0$$

is exact, it follows that M_{i+1} is artinian and noetherian, see Exercise 16.5.

Conversely, let M be a non-zero artinian and noetherian module. Let $M_0 = \{0\}$ and M_1 be minimal among the non-zero submodules of M (it exists by Proposition 16.2). If $M_1 \neq M$, let M_2 be minimal among those submodules of M such that $M_1 \subsetneq M_2$. This procedure produces a sequence

$$\{0\} = M_0 \subsetneq M_1 \subsetneq M_2 \subsetneq \cdots$$

of submodules of M, where each M_{i+1}/M_i is non-zero and admits no proper submodules. Since M is noetherian, the sequence stabilizes and hence it follows that $M_n = M$ for some n. □

17.3. Definition. Let M be a module. We say that the composition series

$$M = V_0 \supseteq V_1 \supsetneq \cdots \supsetneq V_k = \{0\}, \quad M = W_0 \supsetneq W_1 \supsetneq \cdots \supsetneq W_l = \{0\},$$

are **equivalent** if $k = l$ and there exists $\sigma \in \mathbb{S}_k$ such that $V_i/V_{i-1} \simeq W_{\sigma(i)}/W_{\sigma(i)-1}$ for all $i \in \{1, \dots, k\}$.

17.4. Exercise. Find all composition series for the $\mathbb{Z}$-module $\mathbb{Z}/6$.

17.5. Theorem (Jordan–Hölder). *Any two composition series for a module are equivalent.*

Proof. Let M be a module and

$$M = V_0 \supsetneq V_1 \supsetneq \cdots \supsetneq V_k = \{0\}, \quad M = W_0 \supsetneq W_1 \supsetneq \cdots \supsetneq W_l = \{0\},$$

be composition series of M. We claim that these composition series are equivalent. We proceed by induction on k. The case $k = 1$ is trivial, as in this case M has no proper submodules and $M \supseteq \{0\}$ is the only possible composition series for M. So assume the result holds for modules with composition series of length $< k$. If $V_1 = W_1$, then V_1 has composition series of lengths $k-1$ and $l-1$. The inductive hypothesis implies that $k = l$ and we are done. So assume that $V_1 \neq W_1$. Since V_1 and W_1 are submodules of M, the sum $V_1 + W_1$ is also a submodule of M. Moreover, M/V_1 has no non-zero proper submodules and hence $V_1 + W_1 = V$. Then

$$M/V_1 = \frac{V_1 + W_1}{V_1} \simeq \frac{V_1}{V_1 \cap W_1}.$$

Since V_1 has a composition series, V_1 is artinian and noetherian by Theorem 17.2. The submodule $U = V_1 \cap W_1$ is also artinian and noetherian and hence, by Theorem 17.2, admits a composition series

$$U = U_0 \supsetneq U_1 \supsetneq \cdots \supsetneq U_r = \{0\}.$$

Thus $V_1 \supsetneq \cdots \supsetneq V_k = \{0\}$ and $V_1 \supseteq U \supsetneq U_1 \supsetneq \cdots \supsetneq U_r = \{0\}$ are both composition series for V_1. The inductive hypothesis implies that $k-1 = r+1$ and that these composition series are equivalent. Similarly,

$$W_1 \supsetneq W_2 \supsetneq \cdots \supsetneq W_l = \{0\}, \quad W_1 \supsetneq U \supsetneq U_1 \supsetneq \cdots \supsetneq U_r = \{0\},$$

are both composition series for W_1 and hence $l-1 = r+1$ and these composition series are equivalent. Therefore $l = k$ and the proof is completed. □

Jordan–Hölder theorem allows us to define the length of modules that admit a composition series.

17.6. Definition. Let M be a module with a composition series. The **length** $\ell(M)$ of M is defined as the length of any composition series of M.

A module is said to be of finite length if it admits a composition series.

17.7. Exercise. If N and Q are modules with composition series and

$$0 \longrightarrow N \xrightarrow{f} M \xrightarrow{g} Q \longrightarrow 0$$

is an exact sequence of modules, then $\ell(M) = \ell(N) + \ell(Q)$.

17.8. Exercise. If A and B are finite-length submodules of M, then

$$\ell(A+B) + \ell(A \cap B) = \ell(A) + \ell(B).$$

§ 18. Akizuki's theorem

We now prove that if R is a unitary commutative artinian ring, then R is noetherian.

18.1. Exercise. Let R be a unitary commutative ring, I be an ideal of R and M be an R-module such that $I \cdot M = \{0\}$. Prove that if M is finitely generated, then M is a finitely generated (R/I)-module with

$$(r+I) \cdot m = r \cdot m, \quad r \in R, m \in M.$$

Recall that an ideal I of a commutative ring R is said to be **prime** if $xy \in I$ implies that $x \in I$ or $y \in I$.

18.2. Exercise. Let R be an unitary commutative artinian ring.

1) Prove that if R is a domain, then R is a field.
2) Prove that prime ideals of R are maximal.

18.3. Theorem (Akizuki). *Let R be a unitary commutative ring. If R is artinian, then R is noetherian.*

Proof. Assume that the result is not true, so there exists an ideal of R that is not finitely generated. Let X be the set of ideals of R that are not finitely generated. Since $X \neq \emptyset$ and R is artinian, there exists a minimal element $I \in X$. The minimality of I implies that if J is an ideal of R such that $J \subsetneq I$, then J is finitely generated.

Claim. Either $RI = \{0\}$ or $RI = I$.

If not, let $r \in R$ be such that $rI \neq \{0\}$ and $rI \neq I$. Since rI is an ideal of R and $rI \subsetneq I$, the minimality of I implies that rI is finitely generated. Let $f \colon I \to rI$, $x \mapsto rx$. Then f is a surjective module homomorphism. Since $RI \neq \{0\}$, f is non-zero. In particular, $\ker f$ is finitely generated, again by the minimality of I. By the first isomorphism theorem, $I/\ker f \simeq rI$ as R-modules. Since $\ker f$ and $I/\ker f \simeq rI$ are finitely generated, I is finitely generated, a contradiction.

Claim. $M = \{r \in R : rI = \{0\}\}$ is a maximal ideal of R.

Routine calculations show that M is an ideal. Since R is artinian, it is enough to show that M is a prime ideal. Let $rs \in M$. Then $(rs)I = \{0\}$. If $r \notin M$, then $rI \neq \{0\}$. By the previous claim, $rI = I$. Thus

$$\{0\} = (rs)I = s(rI) = sI$$

and hence $s \in M$.

Since M is maximal, $K = R/M$ is a field. Since $MI = \{0\}$, I is an (R/M)-module, that is I is a K-vector space. By Exercise 18.1, $\dim_K I = \infty$. Let B be a basis of I (as a K-vector space) and $x_0 \in B$. Let J be the subspace of I generated by $B \setminus \{x_0\}$. A direct calculation shows that J is an ideal of R. Since $\dim_K J = \infty$, it follows that J is not a finitely generated ideal of R (Exercise 18.1). This is a contradiction, because J is an ideal of R such that $J \subsetneq I$. □

§ 19. Semiprimitive rings

19.1. Definition. A ring R is **semiprimitive** (or Jacobson semisimple) if $J(R) = \{0\}$.

In Lecture 5 we defined primitive rings as those rings that have a faithful simple module. We claim that primitive rings are semiprimitive. If R is primitive, then $\{0\}$ is a primitive ideal. Since $J(R)$ is the intersection of primitive ideals, it follows that $J(R) = \{0\}$.

19.2. Example. If $R = \prod_{i\in I} R_i$ is a direct product of semiprimitive rings, then R is semiprimitive. In fact,

$$J(R) = J\left(\prod_{i\in I} R_i\right) = \prod_{i\in I} J(R_i) = \{0\}.$$

19.3. Example. $\mathbb{Z}$ is semiprimitive, as $J(\mathbb{Z}) = \bigcap_p p\mathbb{Z} = \{0\}$.

19.4. Example. Let $R = C[a,b]$ be the ring of continuous maps $f\colon [a,b] \to \mathbb{R}$. In this case $J(R)$ is the intersection of all maximal ideals of R. Note that each maximal ideal of R is of the form

$$U_c = \{f \in C[a,b] : f(c) = 0\}$$

for some $c \in [a,b]$. Thus $J(R) = \bigcap_{a\leq c\leq b} U_c = \{0\}$.

We proved in Theorem 11.10 (Lecture 8) that $R/J(R)$ is semiprimive.

19.5. Definition. Let $\{R_i : i \in I\}$ be an arbitrary family of rings. For each $j \in I$, let

$$\pi_j\colon \prod_{i\in I} R_i \to R_j$$

be the canonical map. We say that R is a **subdirect product** of $\{R_i : i \in I\}$ if the following conditions hold:

1) There exists an injective ring homomorphism $f\colon R \to \prod_{i\in I} R_i$.
2) For each j, the composition $\pi_j f\colon R \to R_j$ is surjective.

Direct products and direct sums of rings are all examples of subdirect products of rings.

19.6. Exercise. Write (if possible) $\mathbb{Z}$ as a non-trivial subdirect product.

19.7. Example. Let R be a ring, $\{I_j : j\}$ be a collection of ideals of R and

$$f\colon R \to \prod_i R/I_i, \quad r \mapsto (r+I_i)_i.$$

For each i, let $R_i = R/I_i$. Then R is a subdirect product of the R_i if and only if f is injective.

19.8. Theorem. *Let R be a non-zero ring. Then R is semiprimitive if and only if R is isomorphic to a subdirect product of primitive rings.*

Proof. Suppose first that R is semiprimitive and let $\{P_i : i \in I\}$ be the collection of primitive ideals of R. This collection is non-empty, as R is non-zero and semiprimitive. Each R/P_j is primitive and

$$\{0\} = J(R) = \bigcap_{i\in I} P_i.$$

For j let $\lambda_j\colon R \to R/P_j$ and $\pi_j\colon \prod_{i\in I} R/P_i \to R/P_j$ be canonical maps. The ring homomorphism

$$\phi\colon R \to \prod_{i\in I} R/P_i, \quad r \mapsto \{\lambda_i(r) : i \in I\},$$

is injective and satisfies $\pi_j\phi(R) = R/P_j$ for all j.

Assume now that R is isomorphic to a subdirect product of primitive rings R_j and let

$$\varphi\colon R \to \prod_{i\in I} R_i$$

be an injective homomorphism such that $\pi_j(\varphi(R)) = R_j$ for all j. For j let $P_j = \ker \pi_j\varphi$. Since $R/P_j \simeq R_j$, each P_j is a primitive ideal. If $x \in \bigcap_{i\in I} P_i$, then $\varphi(x) = 0$ and thus $x = 0$. Hence $J(R) \subseteq \bigcap_{i\in I} P_i = \{0\}$. □

19.9. Example. The ring $C[a,b]$ of Example 19.4 is isomorphic to a subdirect product of the fields $C[a,b]/U_c \simeq \mathbb{R}$.

§ 20. Jacobson's density theorem

At this point, it is convenient to recall that modules over division rings are pretty much as vector spaces over fields. For that reason, they are often called **vector spaces over division rings**.

20.1. Definition. Let D be a division ring, and V be a vector space over D. A subring $R \subseteq \mathrm{End}_D(V)$ is a **dense ring of linear operators** of V (or simple, **dense** in V) if for every $n \in \mathbb{Z}_{>0}$, every linearly independent set $\{x_1,\dots,x_n\} \subseteq V$ and every (not necessarily linearly independent) subset $\{y_1,\dots,y_n\} \subseteq V$ there exists $f \in R$ such that $f(x_j) = y_j$ for all $j \in \{1,\dots,n\}$.

20.2. Proposition. *Let D be a division ring and V be a finite-dimensional D-vector space. Then $\mathrm{End}_D(V)$ is the only dense ring of V.*

Proof. Let R be dense in V and let $\{x_1,\dots,x_n\}$ be a basis of V. By definition,

$$R \subseteq \mathrm{End}_D(V).$$

If $g \in \mathrm{End}_D(V)$ then, since R is dense in V, there exists $f \in R$ such that $f(x_j) = g(x_j)$ for all $j \in \{1,\dots,n\}$. Hence $g = f \in R$. □

20.3. Theorem (Jacobson). *A unitary ring is primitive if and only if it is isomorphic to a dense ring on a vector space over a division ring.*

Proof. Let R be a unitary ring. If R is isomorphic to a dense ring in V, where V is a D-vector space for some division ring D, then R is primitive, as V is a simple and faithful R-module. Why faithful? If $f \in \operatorname{Ann}_R(V)$, then $f = 0$ since $f(v) = 0$ for all $v \in V$. Why simple? If $W \subseteq V$ is a non-zero submodule, let $v \in V$ and $w \in W \setminus \{0\}$. There exists $f \in R$ such that $v = f(w) \in W$.

Assume now that R is a primitive ring. Let V be a simple faithful module. Schur's lemma implies that $D = \operatorname{End}_R(V)$ is a division ring. Thus V is a D-vector space with

$$D \times V \to V, \quad (\delta, v) \mapsto \delta v = \delta(v).$$

For $r \in R$ let

$$\gamma_r \colon V \to V, \quad v \mapsto rv.$$

A straightforward calculation shows that $\gamma_r \in \operatorname{End}_D(V)$ and that $\gamma\colon R \to \operatorname{End}_D(V)$, $r \mapsto \gamma_r$, is a ring homomorphism. Since V is faithful, $R \simeq \gamma(R) = \{\gamma_r : r \in R\}$. In fact, if $\gamma_r = \gamma_s$, then $rv = \gamma_r(v) = \gamma_s(v) = sv$ for all $v \in V$ and hence $r = s$, as $(r-s)v = 0$ for all $v \in V$.

To prove that R is dense we proceed by induction on n. Assume first that $n = 1$. Since V is simple, $Rx_1 = V$ since $x_1 \neq 0$. Then there exists $r \in R$ such that $rx_1 = y_1$.

Let $n \geq 1$ and assume that the theorem holds for all sets of size $\leq n-1$. Let $\{x_1, \dots, x_n\}$ be a linearly independent subset of V and $\{y_1, \dots, y_n\}$ be a subset of V.

Claim. There exist $\lambda_1, \dots, \lambda_n \in R$ such that $\lambda_i x_i \neq 0$ for all $i \in \{1, \dots, n\}$ and $\lambda_i x_j = 0$ for all $i, j \in \{1, \dots, n\}$ with $i \neq j$.

We first note that the claim implies the theorem. As the theorem holds for $n = 1$, for each $i \in \{1, \dots, n\}$ there exists $r_i \in R$ such that $r_i \lambda_i x_i = y_i$. Let $r = r_1\lambda_1 + \cdots + r_n\lambda_n \in R$. Then

$$rx_i = (r_1\lambda_1 + \cdots + r_n\lambda_n)x_i = r_i\lambda_i x_i = y_i$$

for all $i \in \{1, \dots, n\}$.

It remains to prove the claim. To prove it, suppose that it is not true. Then $\lambda x_i = 0$ for all $i \in \{1, \dots, n-1\}$ implies that $\lambda x_n = 0$. By the inductive hypothesis, given $z_1, \dots, z_{n-1} \in R$, there exists $r \in R$ such that $rx_i = z_i$ for all $i \in \{1, \dots, n-1\}$. This allows us to define a map

$$\phi\colon V^{n-1} \to V, \quad \phi(z_1, \dots, z_{n-1}) = rx_n.$$

The map ϕ is well-defined: if $sx_i = z_i$ for all $i \in \{1, \dots, n-1\}$, then, since $(r-s)x_i = 0$ for all $i \in \{1, \dots, n-1\}$, it follows that $(r-s)x_n = 0$. Hence $rx_n = sx_n$. Moreover, a direct calculation shows that ϕ is an R-module homomorphism.

For $i \in \{1, \dots, n-1\}$, let

$$\xi_i \colon V \xrightarrow{\iota_i} V^{n-1} \xrightarrow{\phi} V,$$

where the map $\iota_i \colon V \to V^{n-1}$ denotes the inclusion in the i-th component. By the inductive hypothesis, for $j \in \{1, \dots, n-1\}$, there exists $r_j \in R$ such that

$$r_j x_i = \begin{cases} x_i & \text{if } i = j, \\ 0 & \text{otherwise.} \end{cases}$$

In particular, for example,

$$\xi_1(x_1) = \phi(x_1, 0, \dots, 0) = \phi(r_1x_1, \dots, r_1x_{n-1}) = r_1x_n.$$

More generally, $\xi_j(x_j) = r_jx_n$ for all $j \in \{1, \dots, n-1\}$. Note that, on the one hand,

$$\phi(x_1, \dots, x_{n-1}) = \phi(r_1x_1, r_2x_2, \dots, r_{n-1}x_{n-1}) = \xi_1x_1 + \cdots + \xi_{n-1}x_{n-1}.$$

On the other hand,

$$x_n = 1x_n = \phi(x_1, \dots, x_{n-1}) = \xi_1x_1 + \cdots + \xi_{n-1}x_{n-1},$$

a contradiction to the linear independence of the set $\{x_1, \dots, x_n\}$. □

20.4. Bonus exercise. Prove Jacobson's density theorem for non-unitary rings.

20.5. Exercise. Let D be a division ring and V be a D-vector space. Prove that if R is dense in V and I is a non-zero ideal of R, then I is dense on V.

20.6. Corollary. *If R is a primitive ring, then either there exists a division ring D such that $R \simeq \operatorname{End}_D(V)$ for some finite-dimensional vector space V over D or for all $m \in \mathbb{Z}_{>0}$ there exists a subring R_m of R and a surjective ring homomorphism $R_m \to \operatorname{End}_D(V_m)$ for some vector space V_m over D such that $\dim_D V_m = m$.*

Proof. The ring R admits a simple faithful module V. Furthermore, by Jacobson's density theorem we may assume that there exists a division ring D such that R is dense in a vector space V over D. Let $\gamma\colon R \to \operatorname{End}_D(V)$, $r \mapsto \gamma_r$, where $\gamma_r(v) = rv$. Since V is faithful, γ is injective. Thus $R \simeq \gamma(R)$.

If $\dim_D V < \infty$, the result follows from Proposition 20.2. Assume that $\dim_D V = \infty$ and let $\{u_1, u_2, \dots\}$ be a linearly independent set. For each $m \in \mathbb{Z}_{>0}$ let V_m be the subspace generated by $\{u_1, \dots, u_m\}$ and $R_m = \{r \in R : rV_m \subseteq V_m\}$. Then R_m is a subring of R. Since R is dense in V, the map

$$R_m \to \operatorname{End}_D(V_m), \quad r \mapsto \gamma_r|_{V_m}$$

is a surjective ring homomorphism. □

§ 21. Prime rings

In commutative algebra, domains play a fundamental role. In non-commutative algebra, certain things could be quite different. For example, the ring $M_n(\mathbb{C})$ is not a domain. We need a non-commutative generalization of domains.

21.1. Definition. Let R be a ring (not necessarily with one). Then R is **prime** if for $x, y \in R$ such that $xRy = \{0\}$ it follows that $x = 0$ or $y = 0$.

A ring R is a **domain** if $xy = 0$ implies $x = 0$ or $y = 0$. Each domain is trivially a prime ring.

21.2. Example. A commutative ring is prime if and only if it is a domain, as $ab = 0$ if and only if $aRb = \{0\}$.

21.3. Example. A non-zero ideal of a prime ring is a prime ring.

21.4. Exercise. A ring is a domain if and only if it is both prime and has no non-zero nilpotent elements.

Rings with no non-zero nilpotent elements are called **reduced**.

A characterization of prime rings:

21.5. Proposition. *Let R be a non-zero ring. The following statements are equivalent:*

1) *R is prime.*

2) *If I and J are left ideals such that $IJ = \{0\}$, then $I = \{0\}$ or $J = \{0\}$.*

3) *If I and J are ideals such that $IJ = \{0\}$, then $I = \{0\}$ or $J = \{0\}$.*

Proof. We first prove that 1) $\implies$ 2). Let I and J be left ideals such that $IJ = \{0\}$. Then $IRJ = I(RJ) \subseteq IJ = \{0\}$. If $J \neq \{0\}$, $u \in I$ and $v \in J \setminus \{0\}$, then $uRv \in IRJ = \{0\}$. Hence $u = 0$.

The implication 2) $\implies$ 3) is trivial.

Let us prove that 3) $\implies$ 1). Let $x, y \in R$ be such that $xRy = \{0\}$. Let $I = RxR$ and $J = RyR$. Since

$$IJ = (RxR)(RyR) \subseteq R(xRy)R = \{0\},$$

we may assume that $I = \{0\}$. In particular, Rx and xR are ideals, as $R(xR) = (Rx)R = \{0\}$. Since $(Rx)(Rx) = \{0\}$, it follows that $Rx = \{0\}$. Similarly, $xR = \{0\}$. Now $\mathbb{Z}x$ is an ideal of R, as

$$R(\mathbb{Z}x) = \mathbb{Z}(Rx) = \{0\} \subseteq \mathbb{Z}x, \quad (\mathbb{Z}x)R = \mathbb{Z}(xR) = \{0\}.$$

Now $(\mathbb{Z}x)R = \{0\}$ implies $\mathbb{Z}x = \{0\}$. Thus $x = 0$. □

Simple rings are trivially prime. The converse is not true. For example, $\mathbb{Z}$ is a domain, so it is a prime ring but is not simple.

21.6. Example. If R_1 and R_2 are rings, $R = R_1 \times R_2$ is not prime, as $I = R_1 \times \{0\}$ and $J = \{0\} \times R_2$ are non-zero ideals such that $IJ = \{0\}$.

A theorem of Connel states (see Theorem **??**) that if K is a field of characteristic zero and G is a group, then $K[G]$ is prime if and only if G does not contain non-trivial finite normal subgroups.

§ 22. Wedderburn's theorem

We studied minimal left ideals of rings before (see Definition 8.14).

22.1. Lemma. *Let R be a prime ring and L be a minimal left ideal of R. Then R is primitive.*

Proof. Since L is a minimal left ideal, it is simple as a module over R. We claim that L is faithful. Let $y \in L \setminus \{0\}$ and $x \in \operatorname{Ann}_R(L)$. Since $xRy \in xRL \subseteq xL = \{0\}$, it follows that $x = 0$. □

22.2. Lemma. *Let D be a division ring and R be a dense ring in a module V over D. If R is left artinian, then $\dim_D V < \infty$.*

Proof. Assume that $\dim_D V = \infty$ and let $\{u_1, u_2, \dots, \}$ be a linearly independent set. Since $R \subseteq \operatorname{End}_D(V)$, it follows that V is a module over R with $f \cdot v = f(v)$, where $f \in R$ y $v \in V$. For $n \in \mathbb{Z}_{>0}$ let

$$I_n = \operatorname{Ann}_R(\{u_1, \dots, u_n\}).$$

Each I_j is a left ideal of R and $I_1 \supseteq I_2 \supseteq \cdots \supseteq I_n \supseteq \cdots$. Let $n \in \mathbb{Z}_{>0}$ and $v \in V \setminus \{0\}$. Since R is dense in V, there exists $f \in R$ such that $f(u_j) = 0$ for all $j \in \{1, \dots, n\}$ and $f(u_{n+1}) = v \neq 0$. Thus $I_1 \supsetneq I_2 \supsetneq \cdots \supsetneq I_n \supsetneq \cdots$, a contradiction. $\square$

22.3. Theorem (Wedderburn). *Let R be a left artinian ring. The following statements are equivalent:*

1) *R is simple.*
2) *R is prime.*
3) *R is primitive.*
4) *$R \simeq M_n(D)$ for some n and some division ring D.*

Proof. The implication 1) $\implies$ 2) is trivial.

To show that 2) $\implies$ 3) first note that R contains a minimal left ideal, as R is left artinian. By Lemma 22.1, R is primitive.

Now we prove that 3) $\implies$ 4). If R is primitive, by Jacobson's density theorem we may assume that there is a division ring D and a D-vector space V such that $R \subseteq \operatorname{End}_D(V)$ is dense in V. Now Lemma 22.2 and Proposition 20.2 imply that $R = \operatorname{End}_D(V) \simeq M_n(D)$, as $\dim_D V < \infty$.

Finally, 4) $\implies$ 1) is trivial, as $M_n(D)$ is simple. $\square$

§ 23. The Artin–Wedderburn theorem

We now prove Artin–Wedderburn theorem. We will assume that our ring is a unitary left artinian ring. One could prove Artin–Wedderburn's theorem for arbitrary rings –see for example [**12**]– but when dealing with unitary rings, the proof is simpler. We will prove that left artinian semiprimitive unitary rings are isomorphic to a direct product of finitely many matrix rings. The idea of the proof goes as follows. We know that if R is semiprimitive, then R is a subdirect product of primitive rings; that is there exists an injective map

$$R \to \prod_{i \in I} R/I_i$$

where each I_i is a primitive ideal. Since R is left artinian, the set I will be finite. Moreover, by Wedderburn's theorem, $R/I_i \simeq M_{n_i}(D_i)$ for some division ring D_i. Finally, a non-commutative version of the Chinese remainder theorem implies that the map is in fact surjective.

23.1. Definition. An ideal I of R is **prime** if $xRy \subseteq I$ implies $x \in I$ or $y \in I$.

Note that a ring R is prime if and only if $\{0\}$ is a prime ideal. Moreover, an ideal I of R is prime if and only if the ring R/I is prime.

23.2. Lemma. *If R is left artinian and I is a primitive ideal, then I is prime.*

Proof. Since I is primitive, then R/I is primitive (Exercise 8.30). By Wedderburn theorem, R/I is prime and hence I is prime. $\square$

23.3. Theorem (Artin–Wedderburn). *Let R be a semiprimitive left artinian unitary ring. Then $R \simeq \prod_{i=1}^{k} M_{n_i}(D_i)$ for finitely many division rings $D_1, \dots, D_k$.*

We shall need the following lemmas.

23.4. Lemma. *Let R be a left artinian ring and I be a primitive ideal. Then I is maximal.*

Proof. If I is a primitive ideal of R, then R/I is a primitive ring by Exercise 8.30. By Wedderburn's theorem, R/I is simple. Thus I is maximal by Proposition 8.19. $\square$

23.5. Lemma. *Let R be a left artinian unitary ring. Let $I_1, \dots, I_k$ be finitely many distinct maximal ideals of R. Then $I_2 \cdots I_k \not\subseteq I_1$.*

Proof. Suppose the result is not true and let k be minimal such that $I_2 \cdots I_k \subseteq I_1$. Since the result is clearly true for two distinct maximal ideals, $k \geq 3$. Let $I = I_2 \cdots I_{k-1}$. Since $I \not\subseteq I_1$, there exists $x \in I \setminus I_1$. Moreover, there exists $y \in I_k \setminus I_1$, as $I_k \neq I_1$. Then $(xR)y \subseteq II_k \subseteq I_1$. Since I_1 is prime (this is left as an exercise), it follows that either $x \in I_1$ or $y \in I_1$, a contradiction. $\square$

23.6. Exercise. Complete the proof of Lemma 23.5.

23.7. Lemma. *Let R be a left artinian unitary ring. Then R has only finitely many primitive ideals.*

Proof. If $I_1, I_2 \dots$ are infinitely many primitive ideals. Since R is left artinian, the sequence $I_1 \supseteq I_1 I_2 \supseteq \cdots$ stabilizes, so there exists n such that

$$I_1 I_2 \cdots I_n = I_1 I_2 \cdots I_n I_{n+1} \subseteq I_{n+1}.$$

This contradicts the previous lemma, as each I_j is a maximal ideal. $\square$

Now we are ready to prove the theorem.

Proof of Theorem 23.3. Let $I_1, \dots, I_k$ be the (distinct) primitive ideals of R. We know that each I_i is a maximal ideal. Thus $I_i + I_j = R$ for $i \neq j$. Since R is semiprimitive, $I_1 \cap \cdots \cap I_k = J(R) = \{0\}$. Let

$$\varphi \colon R \to \prod_{i=1}^{k} R/I_i, \quad x \mapsto (x + I_1, \dots, x + I_k).$$

Then φ is a ring homomorphism with kernel $I_1 \cap \cdots \cap I_k = \{0\}$, so φ is injective. We need to prove that φ is surjective.

We first claim that $I_1 + (I_2 \cdots I_k) = R$. In fact, since $I_1, \dots, I_k$ are maximal ideals, $I_2 \cdots I_k \not\subseteq I_1$. This implies that $I_1 + (I_2 \cdots I_k)$ is an ideal of R that contains I_1. Since I_1 is maximal,

$$I_1 + (I_2 \cdots I_k) = R.$$

Since $I_1 + (I_2 \cdots I_k) = R$, there exists $x_1 \in \prod_{j=2}^{k} I_j$ such that $1 \in x_1 + I_1$. Note that

$$x_1 = (1 + I_1) \cap (I_2 \cdots I_k) \subseteq I_j$$

for all $j \in \{2, \dots, k\}$. Thus

$$\varphi(x_1) = (x_1 + I_1, x_1 + I_2, \dots, x_1 + I_k) = (x_1 + I_1, I_2, \dots, I_k).$$

Similarly, there exist $x_2 \in 1 + I_2, \dots, x_k \in 1 + I_k$ such that

$$\varphi(x_2) = (I_1, x_2 + I_2, \dots, I_k),$$

$$\vdots$$

$$\varphi(x_k) = (I_1, I_2, \dots, x_k + I_k).$$

From this, it follows that φ is surjective: if $(x_1+I_1,\dots,x_k+I_k)\in\prod_{j=1}^k R/I_j$, then

$$\begin{aligned}\varphi(x_1+\cdots+x_k)&=\varphi(x_1)+\cdots+\varphi(x_k)\\&=(x_1+I_1,I_2,\dots,I_k)+\cdots+(I_1,I_2,\dots,x_k+I_k)\\&=(x_1+I_1,\dots,x_k+I_k).\end{aligned}$$

Each R/I_i is primitive and hence isomorphic to $M_{n_i}(D_i)$ for some n_i and some division ring D_i (by Wedderburn's theorem). Therefore

$$R\simeq R/I_1\times\cdots\times R/I_k\simeq\prod_{i=1}^k M_{n_i}(D_i).$$ □

§ 24. Semisimple modules

In the first lectures, we studied semisimple modules over finite-dimensional algebras. Let us now review the theory of semisimple modules over rings. A (finitely generated) module M (over a ring R) is **semisimple** if it is isomorphic to a (finite) direct sum of simple modules.

24.1. Proposition. *Let R be a left artinian ring. Then every non-zero left ideal contains a minimal left ideal.*

Proof. Let I be a non-zero left ideal and X be the family of non-zero left ideals contained in I. Then X is non-empty, as $I\in X$. Then X contains a minimal element by Proposition 16.2. □

24.2. Definition. A ring R with identity is **semisimple** if it is a direct sum of (finitely many) minimal left ideals.

Why finitely many minimal left ideals? Suppose that $R=\oplus_{i\in I}L_i$, where $\{L_i:i\in I\}$ is a collection of minimal left ideals of R. Since R is unitary, $1=\sum_{i\in I}e_i$ (finite sum) for some $e_i\in L_i$. This means that the set $J=\{i\in I:e_i\neq 0\}$ is finite. Note that $R=\oplus_{j\in J}L_j$, as if $x\in R$, then

$$x=x1=\sum_{j\in J}xe_j\in\bigoplus_{j\in J}L_j.$$

Note that ${}_RR$ is finitely generated by $\{1\}$. Minimal left ideals of R are exactly the simple submodules of ${}_RR$. This means that the ring R is semisimple if and only if the module ${}_RR$ is semisimple.

24.3. Proposition. *Let R be a semisimple ring. Then ${}_RR$ is noetherian and artinian.*

Proof. Write R as a direct sum $R=L_1\oplus\cdots\oplus L_n$ of minimal left ideals. Since each L_j is a simple submodule of ${}_RR$, it follows that

$$L_1\oplus\cdots\oplus L_n\supsetneq L_2\oplus\cdots\oplus L_n\supsetneq\cdots\supsetneq L_n\supsetneq\{0\}$$

is a composition series for ${}_RR$ with composition factors $L_1,\dots,L_n$. Since the module ${}_RR$ admits a composition series, it is artinian and noetherian by Theorem 17.2. □

The previous proposition shows that every semisimple ring is both left artinian and left noetherian.

24.4. Exercise. If R is a semisimple ring, every R-module is semisimple.

24.5. Exercise. Prove that if D is a division ring, then $M_n(D)$ is semisimple.

To see a concrete example, note that $M_2(\mathbb{R})$ is semisimple, as

$$M_2(\mathbb{R}) = \left\{\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right\} = \left\{\begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix}\right\} \oplus \left\{\begin{pmatrix} 0 & c \\ 0 & d \end{pmatrix}\right\} \simeq D \oplus D$$

and D is a minimal left ideal of $M_2(\mathbb{R})$.

24.6. Theorem. *Let R be a unitary ring. Then R is semisimple if and only if R is left artinian and $J(R) = \{0\}$.*

Proof. If R is semisimple, then R is left artinian by the previous proposition. Moreover, there are finitely many minimal left ideals $L_1, \dots, L_k$ of R such that $R \simeq L_1 \oplus \cdots \oplus L_k$. We claim that for each $i \in \{1, \dots, k\}$, the left ideal $M_i = \sum_{j \neq i} L_j$ of R is maximal. For example, let us prove that M_1 is maximal. If not, there exists a proper left ideal I of R such that $M_1 \subsetneq I$. Let $x \in I \setminus M_1$ and write

$$x = x_1 + x_2 + \cdots + x_k$$

for $x_j \in L_j$. Since $x_2 + \cdots + x_k \in M_1 \subseteq I$, it follows that $x_1 \in I \cap L_1$. Thus $I \cap L_1$ is a non-zero left ideal contained in the minimal left ideal L_1, a contradiction. Now the claim follows, as $J(R) \subseteq M_1 \cap \cdots \cap M_k = \{0\}$.

Conversely, if R is left artinian and $J(R) = \{0\}$, then $R \simeq M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$ for division rings $D_1, \dots, D_k$, this is the Artin–Wedderburn theorem. Since each $M_{n_j}(D_j)$ is semisimple, it follows that R is semisimple. $\square$

§ 25. The Hopkins–Levitzki theorem

25.1. Theorem (Hopkins–Levitzki). *Let R be a unitary left artinian ring. Then R is left noetherian.*

Proof. Let $J = J(R)$. Since R is left artinian, J is a nilpotent ideal by Theorem 16.9. Let n be such that $J^n = \{0\}$. Now consider the sequence

$$R \supsetneq J \supseteq J^2 \supseteq \cdots \supseteq J^{n-1} \supseteq J^n = \{0\}.$$

Each J^i/J^{i+1} is a module over R annihilated by J, that is $J \cdot (J^i/J^{i+1}) = \{0\}$, as

$$x \cdot (y + J^{i+1}) = xy + J^{i+1} \subseteq JJ^i + J^{i+1} = J^{i+1}$$

if $x \in J$ and $y \in J^i$. Thus each J^i/J^{i+1} is a module over R/J. Since R/J is left artinian and $J(R/J) = \{0\}$ by Theorem 11.10, it follows that R/J is semisimple. In particular, since every (R/J)-module is semisimple, each J^i/J^{i+1} is semisimple and hence it is left noetherian.

Now suppose that R is not left noetherian. Let m be the largest non-negative integer such that J^m is not left noetherian. Note that $0 \leq m < n$. The sequence

$$0 \longrightarrow J^{m+1} \longrightarrow J^m \longrightarrow J^m/J^{m+1} \longrightarrow 0$$

is exact. Since J^{m+1} is left noetherian by the definition of m and J^m/J^{m+1} is left noetherian, it follows that J^m is noetherian, a contradiction. $\square$

§ 26. Local rings

In this section, we will consider arbitrary rings with one.

26.1. Definition. A ring is said to be **local** if it contains only one maximal left ideal.

Division rings are local rings.

26.2. Theorem. *Let R be a ring and $I = R \setminus \mathcal{U}(R)$. The following statements are equivalent:*

1) *R is local.*
2) *$R/J(R)$ is a division ring.*
3) *$I = J(R)$.*
4) *I is an ideal of R.*

Proof. We first prove 1) $\implies$ 2). Let M be the maximal left ideal of R. Then $J(R) = M$. Let $x \notin M$. Then $R = Rx + M$, so $1 = rx + m$ for some $r \in R$ and $m \in M$. Thus $r + M$ is a left inverse of $x + M$. In particular, $r \notin M$. Since $R = Rr + M$, there exists $y \in R$ such that $1 = yr$. Therefore $y + M$ is a left inverse of $r + M$. Thus

$$\begin{aligned} y + M &= (y + M)(1 + M) = (y + M)(r + M)(x + M) \\ &= (yr + M)(x + M) = (1 + M)(x + M) = x + M \end{aligned}$$

and hence $x + M$ is invertible.

Now we prove 2) $\implies$ 3). Clearly $J(R) \subseteq I$.

Conversely, let $x \in I$. If $x \notin J(R)$, then $x + J(R) \neq J(R)$. Since $R/J(R)$ is a division ring, $x + J(R) \in \mathcal{U}(R/J(R))$. In particular, $1 - xy \in J(R)$ and hence $xy = 1 - (1 - xy) \in \mathcal{U}(R)$. Thus $1 = (xy)z = x(yz)$ for some $z \in R$ and therefore $x \notin I$, a contradiction.

It is trivial that 3) $\implies$ 4).

Finally, we prove 4) $\implies$ 1). Let M be a maximal left ideal of R. Then $M \subseteq I$. Since M is maximal and I is in particular a left ideal of R, it follows that $M = I$. □

26.3. Definition. An element x of a ring is said to be **idempotent** if $x^2 = x$.

Examples of idempotents are 0 and 1. An idempotent x is said to be **non-trivial** if $x \notin \{0, 1\}$.

26.4. Exercise. Let p be a prime number and $m > 0$. Prove that the only idempotents of $\mathbb{Z}/p^m$ are 0 and 1.

26.5. Exercise. How many idempotents does $\mathbb{Z}/n$ have?

26.6. Exercise. Let R be a ring with one and I be an ideal of R. We say that an idempotent $x \in R/I$ can be lifted if $x = e + I$ for some idempotent e of R. Prove that if every element of I is nilpotent, then every idempotent of R/I can be lifted.

The previous exercise shows that if R is left artinian, every idempotent of $R/J(R)$ can be lifted to R.

26.7. Lemma. *Let R be a left artinian ring. Then $J(R)$ is nil.*

Proof. Let $x \in J(R)$. The sequence $Rx \supseteq Rx^2 \supseteq \cdots$ stabilizes, so $Rx^n = Rx^{n+1}$ for some n. In particular, there exists $r \in R$ such that $x^n = rx^{n+1}$. This implies that $(1-rx)x^n = 0$. Since $x \in J(R)$, the element $1-rx$ is invertible. Hence $x^n = 0$. □

26.8. Theorem. *Let R be a left artinian ring. Then R is local if and only if R has no non-trivial idempotents.*

Proof. Let us first prove $\implies$. For this implication, we do not need to use that R is left artinian. Let $x \in R$ be an idempotent. Then $x(1-x) = 0$. If $x \in \mathcal{U}(R)$, then $x = 1$. If $1-x \in \mathcal{U}(R)$, then $x = 0$. If $x \notin \mathcal{U}(R)$ and $1-x \notin \mathcal{U}(R)$, then, since $R \setminus \mathcal{U}(R)$ is an ideal of R, it follows that $1 = x+1-x \notin \mathcal{U}(R)$, a contradiction.

Now we prove $\impliedby$. By the previous lemma, $J(R)$ is nil. By the previous exercise, every idempotent of $R/J(R)$ can be lifted. Thus $R/J(R)$ has no non-trivial idempotents. On the other hand, by Artin–Wedderburn theorem,

$$R/J(R) \simeq \prod_{i=1}^{k} M_{n_i}(D_i)$$

for some $n_1, \dots, n_k \geq 1$ and division rings $D_1, \dots, D_k$. Then $k = n_1 = 1$, as $R/J(R)$ has no non-trivial idempotents. Since $R/J(R)$ is a division ring, R is local by the previous theorem. □

26.9. Theorem. *The center of a local ring is local.*

Proof. Let R be a local ring. By Theorem 26.2, $J(R) = R \setminus \mathcal{U}(R)$. We need to prove that $Z(R) \setminus \mathcal{U}(Z(R)) = J(Z(R))$. We first note that

$$\mathcal{U}(Z(R)) = Z(R) \cap \mathcal{U}(R). \tag{26.1}$$

We claim that $Z(R) \cap J(R) \subseteq J(Z(R))$. Let $x \in Z(R) \cap J(R)$. Let $z \in Z(R)$. Since $x \in J(R)$, $1-zx \in \mathcal{U}(R)$. Moreover, $1-zx \in Z(R)$. Thus

$$1-zx \in Z(R) \cap \mathcal{U}(R) = \mathcal{U}(Z(R)).$$

Hence $x \in J(Z(R))$.

To prove the theorem it is enough to show that $Z(R) \setminus \mathcal{U}(Z(R)) = J(Z(R))$. Let us prove the non-trivial inclusion. Let $x \in Z(R) \setminus \mathcal{U}(Z(R))$. Then (26.1) implies that $x \notin \mathcal{U}(R)$. By Theorem 26.2, $x \in J(R)$. Then $x \in J(R) \cap Z(R) \subseteq J(Z(R))$. □

26.10. Exercise. Let R be a local ring. Prove that $Z(R) \cap J(R) = J(Z(R))$.

26.11. Exercise. Prove that a ring is local if and only if it contains only one maximal right ideal.

26.12. Exercise. Find a non-local ring with a unique maximal ideal.

26.13. Exercise. Let R be a ring with at least three elements. If $|\mathcal{U}(R)| = 1$, then R is not local.

26.14. Exercise. A ring R is said to be **Von Neumann regular** if for every non-zero $r \in R$, $r = rxr$ for some $x \in R$. Prove that a local Von Neumann ring is a division ring.

26.15. Exercise. Let R be a ring such that every element of R is either nilpotent or a unit. Prove that R is local.

26.16. Bonus exercise. A ring R is said to be **semilocal** if $R/J(R)$ is left artinian. Prove the following statements:

1) Every local ring is semilocal.
2) R is semilocal if and only if $R/J(R)$ is semisimple.
3) If R has finitely many maximal left ideals, then R is semilocal.
4) If $R_1, \dots, R_k$ are rings, then $\bigoplus_{i=1}^{k} R_i$ is semilocal if and only if each R_i is semilocal.

26.17. Bonus exercise. Let R be a ring such that $R/J(R)$ is commutative. Prove that R is semilocal if and only if R has finitely many maximal ideals.

§ 27. Gustafson's theorem

In 1905, Schur proved that the maximum number of linearly independent $n \times n$ complex matrices is $\lfloor n^2/4 \rfloor + 1$, where $x \mapsto \lfloor x \rfloor$ is the greatest integer function. We will prove Schur's theorem by representation-theoretic methods.

27.1. Theorem (Gustafson). *Let R be a complex commutative finite-dimensional algebra and M be a faithful n-dimensional R-module. Then*

$$\dim_{\mathbb{C}} R \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 1.$$

Recall that the **floor function** is the function that takes an element $x \in \mathbb{R}$ and returns

$$\lfloor x \rfloor = \max\{m \in \mathbb{Z} : m \leq x\}.$$

Thus $x - 1 < \lfloor x \rfloor \leq x$.

27.2. Exercise. For $n \geq 1$, let $a_n = \lfloor n^2/4 \rfloor + 1$. Then

$$a_{n_1+\cdots+n_k} \geq a_{n_1} + \cdots + a_{n_k}.$$

We now need some exercises on commutative artinian unitary rings.

27.3. Exercise. Let R be a commutative artinian unitary ring. Prove that R has finitely many maximal ideals.

27.4. Lemma. *Let R be a commutative artinian unitary ring. Then $R \simeq R_1 \times \cdots \times R_k$ for local rings $R_1, \dots, R_k$.*

Proof. By Lemma 23.7, R has finitely many primitive ideals, say $I_1, \dots, I_k$. Each I_j is a maximal ideal by Lemma 23.4. Moreover, $J(R) = I_1 \cap \cdots \cap I_k$. Since R is artinian, $J(R)$ is nilpotent by Theorem 16.9. Thus

$$\{0\} = J(R)^m = (I_1 \cap \cdots \cap I_k)^m = I_1^m \cap \cdots \cap I_k^m$$

for some $m \geq 1$. By the Chinese remainder theorem,

$$R \simeq R/(I_1^m \cdots I_k^m) \simeq \prod_{j=1}^{k} R/I_j^m.$$

For $j \in \{1, \dots, k\}$, let $R_j = R/I_j^m$ and $\pi_j \colon R \to R_j$ be the canonical map. We claim that each R_j is a local ring with maximal ideal $\pi(I_j)$. By the correspondence theorem, the maximal ideals of R_j are in bijective correspondence with the maximal ideals of R containing I_j^m. Let M be a maximal ideal of R_j. We want to prove that $M = \pi(I_j)$. If $x \in I_j$, then $x^m \in I_j^m \subseteq \pi^{-1}(M)$. Since $\pi^{-1}(M)$ is a maximal ideal, it is a prime ideal of R. Thus $x \in \pi^{-1}(M)$, that is $\pi(x) \in M$. Since $\pi(I_j) \subseteq M$ and $\pi(I_j)$ is a maximal ideal of R, it follows that $\pi(I_j) = M$. □

We also need the following very useful lemma:

27.5. Lemma (Nakayama). *Let R be a unitary ring and M be a finitely generated R-module. If $J(R) \cdot M = M$, then $M = \{0\}$.*

Proof. Assume that M is generated by $x_1, \dots, x_n$. Since $x_n \in M = J(R) \cdot M$, there exist $r_1, \dots, r_n \in J(R)$ such that $x_n = r_1 \cdot x_1 + \cdots + r_n \cdot x_n$, that is $(1 - r_n) \cdot x_n = \sum_{j=1}^{n-1} r_j \cdot x_j$. Since $1 - r_n$ is invertible, there exists $s \in R$ such that $s(1 - r_n) = 1$. Then $x_n = \sum_{j=1}^{n-1} (sr_j) \cdot x_j$ and hence M is generated by $x_1, \dots, x_{n-1}$. After repeating this procedure sufficiently many times (finitely many times), one gets $M = \{0\}$. □

27.6. Exercise. Let R be a unitary ring and M be a finitely generated module.

1) If N be a submodule of M such that $J(R) \cdot M/N = M/N$, then $M = N$.
2) If $\pi \colon M \to M/(J(R) \cdot M)$ be the canonical map and $M/(J(R) \cdot M)$ is generated by $\{\pi(m_1), \dots, \pi(m_k)\}$ for some $m_1, \dots, m_k \in M$, then M is generated by $\{m_1, \dots, m_k\}$.

Recall that an idempotent of a ring R is an element $e \in R$ such that $e^2 = e$. Two idempotents e and f are said to be **orthogonal** if $ef = fe = 0$.

27.7. Exercise. Let R be a unitary ring. Prove that the following statements are equivalent:

1) $R = I_1 \oplus \ldots \oplus I_n$ as a direct sum of left ideals $I_1, \dots, I_n$ of R.
2) $1 = e_1 + \cdots + e_n$ for some orthogonal idempotents $e_1, \dots, e_n$.

We are ready to prove Gustafson's theorem.

Proof of Theorem 27.1. Assume first that R is a local ring and let I be its maximal ideal. Then $F = R/I$ is a field. Note that F is a finite extension of $\mathbb{C}$, so it is an algebraic extension. Since $\mathbb{C}$ is algebraically closed, $F \simeq \mathbb{C}$. Moreover, $M/(I \cdot M)$ is a vector space over $\mathbb{C}$. Let $k \in \mathbb{Z}$ be such that $k = \dim(I \cdot M)$. Then $\dim M/(I \cdot M) = n - k$. Let

$$\{m_1 + I \cdot M, \dots, m_{n-k} + I \cdot M\}$$

be a basis of $M/(I \cdot M)$. Since $I = J(R)$, it follows from Exercise 27.6 that M is generated by $\{m_1, \dots, m_{n-k}\}$. Let V be the complex vector space generated by $m_1, \dots, m_{n-k}$ and W

be the set of complex linear maps $V \to I \cdot M$. Then

$$\dim_{\mathbb{C}} W = \dim_{\mathbb{C}}(I \cdot M) \dim_{\mathbb{C}} V \leq k(n-k).$$

Let $f\colon I \to W$, $r \mapsto (v \mapsto r \cdot v)$. Then f is a well-defined complex linear map.

We claim that f is injective: if $r \in \ker f$, then $f(r) = 0$. Thus $r \cdot v = 0$ for all $v \in V$. In particular, since $r \cdot m_i = 0$ for all $i \in \{1, \dots, n-k\}$, it follows that $r \cdot m = 0$ for all $m \in M$. Since M is faithful, $r = 0$.

Since f is injective,

$$\dim_{\mathbb{C}} I \leq \dim_{\mathbb{C}} W \leq k(n-k) = \frac{n^2}{4} - \left(\frac{n}{2} - k\right)^2 \leq \frac{n^2}{4}.$$

Hence $\dim_{\mathbb{C}} I \leq \lfloor n^2/4 \rfloor$. Now

$$\dim_{\mathbb{C}} R = \dim_{\mathbb{C}} I + \dim_{\mathbb{C}}(R/I) \leq \left\lfloor n^2/4 \right\rfloor + 1.$$

Assume now that R is a commutative artinian ring. By Lemma 27.4, we may assume that $R = \bigoplus_{i=1}^{k} R_i$, where $R_1, \dots, R_k$ are local artinian rings and ideals of R. Since R is unitary, $1 = e_1 + \cdots + e_k$ for orthogonal idempotents $e_1 \in R_1, \dots, e_k \in R_k$ (see Exercise 27.7). In particular, $M = \sum_{j=1}^{k} e_j \cdot M$.

We claim that $M = \bigoplus_{j=1}^{k} e_j \cdot M$. For example, let $x \in e_1 \cdot M \cap (e_2 \cdot M + \cdots + e_k \cdot M)$. Then $x = e_1 \cdot m_1 = e_2 \cdot m_2 + \cdots + e_k \cdot m_k$ for some $m_1, \dots, m_k \in M$. Since $e_1^2 = e_1$ and $e_1 e_j = 0$ for all $j \in \{2, \dots, k\}$, we obtain that

$$\begin{aligned} e_1 \cdot x &= e_1^2 \cdot m_1 = e_1 \cdot m_1 = x, \\ e_1 \cdot x &= (e_1 e_2) \cdot m_2 + \cdots + (e_1 e_k) \cdot m_k = 0. \end{aligned}$$

Then $x = 0$. In this way we prove that $M = \bigoplus_{j=1}^{k} e_j \cdot M$.

Each $e_j \cdot M$ is a faithful R_j-module. It is trivial to see that each $e_j \cdot M$ is an R_j-module. Let us show that, for example, $e_1 \cdot M$ is faithful. Let $r \in \operatorname{Ann}_{R_1}(e_1 \cdot M)$. Then

$$(re_1) \cdot m = r \cdot (e_1 \cdot m) = 0$$

for all $m \in M$. Thus $re_1 = 0$, as $re_1 \in \operatorname{Ann}_R(M) = \{0\}$ since M is faithful. Moreover,

$$r = r1 = r(e_1 + \cdots + e_k) = re_1 + re_2 + \cdots + re_k = re_2 + \cdots + re_k.$$

Thus $r \in R_1 \cap (R_2 + \cdots + R_k) = \{0\}$. Similarly one proves that each $e_j \cdot M$ is a faithful R_j-module. For $j \in \{1, \dots, k\}$, let $n_j = \dim_{\mathbb{C}} e_j \cdot M$. Then

$$n = \dim_{\mathbb{C}} M = \sum_{j=1}^{k} \dim_{\mathbb{C}}(e_j \cdot M) = \sum_{j=1}^{k} n_j.$$

Using Exercise 27.2 and the previous case,

$$\begin{aligned} \dim_{\mathbb{C}} R &= \sum_{j=1}^{k} \dim_{\mathbb{C}} R_j \leq \sum_{j=1}^{k} \left(\left\lfloor \frac{n_j^2}{4} \right\rfloor + 1 \right) \\ &\leq \left\lfloor \frac{(n_1 + \cdots + n_k)^2}{4} \right\rfloor + 1 = \left\lfloor \frac{n^2}{4} \right\rfloor + 1. \end{aligned}$$

$\square$

The proof of Gustafson's theorem we presented can be easily adapted for arbitrary algebraically closed fields. Using the technique of **extensions of scalars** (see Theorem 33.17), we can get the result for other fields. Let K be a field and $\overline{K}$ be its algebraic closure. If R is a finite-dimensional commutative K-algebra and M is a faithful R-module with $\dim_K M = n < \infty$, then $\overline{R} = R \otimes_K \overline{K}$ is a commutative $\overline{K}$-algebra and $\dim_{\overline{K}} \overline{R} = \dim_K R$. Moreover, $\overline{M} = M \otimes_K \overline{K}$ is a faithful $\overline{R}$-module with $\dim_{\overline{K}} \overline{M} = \dim_K M$. By Gustafson's theorem,

$$\dim_K R = \dim_{\overline{K}} \overline{R} \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 1.$$

§ 28. Schur's theorem

We now present an application. For that purpose, we need a lemma.

28.1. Lemma. *Let $\{A_1, \dots, A_m\} \subseteq M_n(\mathbb{C})$ be a maximal linearly independent set of commuting matrices. If R is the complex vector space generated by $\{A_1, \dots, A_m\}$, then R is a commutative finite-dimensional complex algebra and $\mathbb{C}^n$ is a faithful module.*

Proof. Let $B \in M_n(\mathbb{C})$ be such that $BA_i = A_iB$ for all $i \in \{1, \dots, m\}$. The set $\{A_1, \dots, A_m, B\}$ is linearly dependent, by the maximality of $\{A_1, \dots, A_m\}$. There exist $\mu, \lambda_1, \dots, \lambda_m \in \mathbb{C}$ not all zero such that

$$\lambda_1 A_1 + \cdots + \lambda_m A_m + \mu B = 0.$$

Taking $B = A_iA_j$ for $i, j \in \{1, \dots, m\}$ we see that all products of the form A_iA_j belong to R. Moreover, taking $B = I$ (the $n \times n$ identity matrix), we see that $I \in R$. A direct calculation now shows that R is a commutative finite-dimensional complex algebra.

To see that $\mathbb{C}^n$ is a faithful R-module, let

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \in \operatorname{Ann}_R(\mathbb{C}^n).$$

Then $Av = 0$ for all $v \in \mathbb{C}^n$. In particular,

$$\begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{n1} \end{pmatrix} = A \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Thus the first column of Av is zero. Similarly one proves that all the other columns of A need to be zero. □

28.2. Theorem (Schur). *Let S be a linearly independent set of commuting matrices of $M_n(\mathbb{C})$. Then*

$$|S| \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 1.$$

Proof. Let $T = \{A_1, \dots, A_m\}$ be a maximal set of linearly independent commuting $n \times n$ complex matrices. Let R be the complex vector space generated by $A_1, \dots, A_m$. By

Lemma 28.1, R is a finite-dimensional commutative complex algebra and $M = \mathbb{C}^n$ is a faithful R-module. Then

$$|S| \leq |T| = m = \dim_{\mathbb{C}} R \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 1,$$

by Gustafson's theorem. □

There are other proofs of Schur's inequality. For example, Mirzakhani found a short and elementary proof, see [**17**].

§ 29. Rickart's theorem

We now consider Jacobson's semisimplicity problem.

29.1. Question. Let G be a group and K be a field. When $J(K[G]) = \{0\}$?

As an application of Amitsur's theorem 12.3, we prove that complex group algebras have null Jacobson radical. This is known as Rickart's theorem. The original proof found by Rickart uses complex analysis. Here, however, we present an algebraic proof.

29.2. Theorem (Rickart). *Let G be a group. Then $J(\mathbb{C}[G]) = \{0\}$.*

To prove the theorem, we need a lemma.

29.3. Lemma. *Let G be a group. Then $J(\mathbb{C}[G])$ is nil.*

Proof. We need to show that every element of $J(\mathbb{C}[G])$ is nilpotent. If G is countable, then the result follows from Amitsur's theorem 12.3. So assume that G is not countable. Let $\alpha \in J(\mathbb{C}[G])$, say

$$\alpha = \sum_{i=1}^{n} \lambda_i g_i,$$

where $\lambda_1, \dots, \lambda_n \in \mathbb{C}$ and $g_1, \dots, g_n \in G$. Let $H = \langle g_1, \dots, g_n \rangle$. Then $\alpha \in \mathbb{C}[H]$ and H is countable. We claim that $\alpha \in J(\mathbb{C}[H])$. Decompose G as a disjoint union

$$G = \bigcup_{\lambda} x_\lambda H$$

of cosets of H in G. Then $\mathbb{C}[G] = \bigoplus_\lambda x_\lambda \mathbb{C}[H]$ and hence $\mathbb{C}[G] = \mathbb{C}[H] \oplus K$ for some right module K over $\mathbb{C}[H]$ (this follows from the fact that one of the cosets is that of H). Since $\alpha \in J(\mathbb{C}[G])$, for each $\beta \in \mathbb{C}[H]$ there exists $\gamma \in \mathbb{C}[G]$ such that $\gamma(1 - \beta\alpha) = 1$. Write $\gamma = \gamma_1 + \kappa$ for $\gamma_1 \in \mathbb{C}[H]$ and $\kappa \in K$. Then

$$1 = \gamma(1 - \beta\alpha) = \gamma_1(1 - \beta\alpha) + \kappa(1 - \beta\alpha)$$

and hence $\kappa(1 - \beta\alpha) \in K \cap \mathbb{C}[H] = \{0\}$, as $\beta \in \mathbb{C}[H]$. Since $1 = \gamma_1(1 - \beta\alpha)$, it follows that $\alpha \in J(\mathbb{C}[H])$ and the lemma follows from Amitsur's theorem 12.3. □

We now prove the theorem.

Proof of Theorem 29.2. For $\alpha = \sum_{i=1}^{n} \lambda_i g_i \in \mathbb{C}[G]$ let

$$\alpha^* = \sum_{i=1}^{n} \overline{\lambda_i} g_i^{-1}.$$

Then $\alpha\alpha^* = 0$ if and only if $\alpha = 0$ and, moreover, $(\alpha\beta)^* = \beta^*\alpha^*$ for all $\beta \in \mathbb{C}[G]$. Assume that $J(\mathbb{C}[G]) \neq \{0\}$ and let $\alpha \in J(\mathbb{C}[G]) \setminus \{0\}$. Then $\beta = \alpha\alpha^* \in J(\mathbb{C}[G])$, as $J(\mathbb{C}[G])$ is an ideal of $\mathbb{C}[G]$. Moreover, the previous lemma implies that β is nilpotent. Note that $\beta \neq 0$, as $\alpha \neq 0$. Since $\beta^* = \beta$,

$$(\beta^m)^* = (\beta^*)^m = \beta^m$$

for all $m \geq 1$. If there exists $k \geq 2$ such that $\beta^k = 0$ and $\beta^{k-1} \neq 0$, then

$$\beta^{k-1}\left(\beta^{k-1}\right)^* = \beta^{2k-2} = 0$$

and hence $\beta^{k-1} = 0$, a contradiction. Thus $\beta = 0$ and therefore $\alpha = 0$. □

29.4. Exercise. If G is a group, then $J(\mathbb{R}[G]) = 0$.

29.5. Definition. A ring R **semiprime** if $aRa = \{0\}$ implies $a = 0$.

29.6. Proposition. *Let R be a ring. The following statements are equivalent:*

1) *R is semiprime.*
2) *If I is a left ideal such that $I^2 = \{0\}$, then $I = \{0\}$.*
3) *If I is an ideal such that $I^2 = \{0\}$, then $I = \{0\}$.*
4) *R does not contain non-zero nilpotent ideals.*

Proof. We first prove that 1) $\implies$ 2). If $I^2 = \{0\}$ y $x \in I$, then $xRx \subseteq I^2 = \{0\}$ and thus $x = 0$.

The implications 2) $\implies$ 3) and 4) $\implies$ 3) are both trivial.

Let us prove that 3) $\implies$ 4). If I is a non-zero nilpotent ideal, let $n \in \mathbb{Z}_{>0}$ be minimal such that $I^n = \{0\}$. Since $(I^{n-1})^2 = \{0\}$, it follows that $I^{n-1} = \{0\}$, a contradiction.

Finally, we prove that 3) $\implies$ 1). Let $a \in R$ be such that $aRa = \{0\}$. Then $I = RaR$ is an ideal of R such that $I^2 = \{0\}$. Thus $RaR = \{0\}$. This means that Ra and aR are ideals such that $(Ra)R = R(aR) = \{0\}$ (for example, $R(aR) \subseteq RaR = \{0\} \subseteq aR$). Moreover, since $(Ra)(Ra) = \{0\}$ and $(aR)(aR) = \{0\}$, it follows that $aR = Ra = \{0\}$. This implies that $\mathbb{Z}a$ is an ideal of R, as $R(\mathbb{Z}a) \subseteq \mathbb{Z}(Ra) = \{0\}$ and $(\mathbb{Z}a)R \subseteq aR = \{0\}$. Now $(\mathbb{Z}a)(\mathbb{Z}a) \subseteq (\mathbb{Z}a)R = \{0\}$ and hence $a = 0$, as $\mathbb{Z}a = \{0\}$. □

Two consequences:

29.7. Exercise. A commutative ring is semiprime if and only if it does not contain non-zero nilpotent elements.

29.8. Exercise. Let D be a division ring.

1) $D[X]$ is semiprime and semiprimitive.
2) $D[\![X]\!]$ is semiprime and it is not semiprimitive.

29.9. Corollary. *The ring $\mathbb{C}[G]$ is semiprime.*

Proof. Since $J(\mathbb{C}[G]) = \{0\}$ by Rickart's theorem and the Jacobson radical contains every nil ideal by Proposition 9.6, it follows that $\mathbb{C}[G]$ does not contain non-trivial nil ideals. Thus $\mathbb{C}[G]$ does not contain non-trivial nilpotent ideals and hence $\mathbb{C}[G]$ is semiprime. □

29.10. Exercise. Prove that $Z(\mathbb{C}[G])$ is semiprime.

We now characterize when complex group algebras are left artinian. For that purpose, we need a lemma. This is similar to one of the implications proved in Proposition 1.23. However, in the arbitrary setting we are considering, we need to use Zorn's lemma.

29.11. Lemma. *Let M be a semisimple module and N be a submodule. Then N is a direct summand.*

Sketch of the proof. Let $M=\oplus_{i\in I}M_i$ be a direct sum of simple submodules. Since each $N\cap M_i$ is a submodule of M_i and M_i is simple, it follows that $N\cap M_i=\{0\}$ or $N\cap M_i=M_i$. If $N\cap M_i=M_i$ for all $i\in I$, then $N=M$ and the lemma is proved. So we may assume that there exists $i\in I$ such that $N\cap M_i=\{0\}$. Let X be the set of subsets J of I such that $N\cap(\oplus_{j\in J}M_j)=\{0\}$. Our assumptions imply that X is non-empty. Zorn's lemma implies the existence of a maximal element K. Let $N_1=\oplus_{k\in K}M_k$. We claim that $N\bigoplus N_1=M$. If not, there exists $i\in I$ such that $M_i\not\subseteq N\oplus N_1$. The simplicity of M_i implies that $M_i\cap(N\oplus N_1)=\{0\}$, which contradicts the maximality of K. □

A direct application of the lemma proves that complex group algebras of infinite groups are never semisimple.

29.12. Proposition. *If G is an infinite group, then $\mathbb{C}[G]$ is not semisimple.*

Proof. Assume that $R=\mathbb{C}[G]$ is semisimple. Let I be the augmentation ideal of R, that is

$$I=\left\{\alpha=\sum_{g\in G}\lambda_g g\in R:\sum_{g\in G}\lambda_g=0\right\}.$$

By Lemma 29.11, there exists a non-zero left ideal J such that $R=I\oplus J$. Since R is unitary, there exist $e\in I$ and $f\in J$ such that $1=e+f$. If $x\in I$, then $x=xe+xf$ and hence $xf=x-xe\in I\cap J=\{0\}$. Since $x=xe$ for all $x\in I$, it follows that $e=e^2$. Similarly, one proves that $f^2=f$. Moreover, $ef=0$, as $ef\in I\cap J=\{0\}$. Since I is the augmentation ideal of R and $If=(Re)f=R(ef)=\{0\}$ (note that $I=Re$ because $x=xe$ for all $x\in I$), we conclude that $(g-1_G)f=0$ for all $g\in G$, as $g-1_G\in I$ for all $g\in G$. If $f=\sum_{h\in G}\lambda_h h$ (finite sum) and $g\in G$, then

$$f=gf=\sum_{h\in G}\lambda_h(gh)=\sum_{h\in G}\lambda_{g^{-1}h}h.$$

Thus $\lambda_h=\lambda_{g^{-1}h}$ for all $g,h\in G$. Since G is infinite, some $\lambda_g=0$ and hence $f=0$. Thus $e=1$ and $I=\mathbb{C}[G]$, a contradiction. □

29.13. Theorem. *Let G be a group. Then $\mathbb{C}[G]$ is left artinian if and only if G is finite.*

Proof. If G is finite, then $\mathbb{C}[G]$ is left artinian because $\dim\mathbb{C}[G]=|G|<\infty$. So assume that G is infinite. By Rickart's theorem, $J(\mathbb{C}[G])=0$. Moreover, $\mathbb{C}[G]$ is not semisimple by the previous proposition. Thus $\mathbb{C}[G]$ is not left artinian by Theorem 24.6. □

§ 30. Maschke's theorem

We now present another instance of the semisimplicity problem. In this case, our result is for finite groups.

30.1. Theorem (Maschke). *Let G be a finite group. Then $J(K[G])=\{0\}$ if and only if the characteristic of K is zero or does not divide the order of G.*

Proof. Assume that $G=\{g_1,\dots,g_n\}$, where $g_1=1$. Let
$$\rho\colon K[G]\to K,\quad \alpha\mapsto \operatorname{trace}(L_\alpha),$$
where $L_\alpha(\beta)=\alpha\beta$. Then
$$\rho(g_i)=\begin{cases}n & \text{if } i=1,\\ 0 & \text{if } 2\leq i\leq n,\end{cases}$$
as $L_{g_i}(g_j)=g_ig_j\neq g_j$ if $i\neq j$ and hence the matrix of L_{g_i} in the basis $\{g_1,\dots,g_n\}$ contains zeros in the main diagonal.

Assume that $J=J(K[G])$ is non-zero and let $\alpha=\sum_{i=1}^n\lambda_ig_i\in J\setminus\{0\}$. Without loss of generality we may assume that $\lambda_1\neq 0$ (if $\lambda_1=0$ there exists some $\lambda_i\neq 0$ and we need to take $g_i^{-1}\alpha\in J$). Then
$$\rho(\alpha)=\sum_{i=1}^n\lambda_i\rho(g_i)=n\lambda_1.$$
Since G is finite, $K[G]$ is a finite-dimensional algebra and hence $K[G]$ is left artinian. Since J is a nilpotent ideal, in particular, α is a nilpotent element. Then L_α is nilpotent and hence $0=\rho(\alpha)=n\lambda_1$. This implies that the characteristic of the field K divides n.

Conversely, let K be a field of prime characteristic and that this prime divides n. Let $\alpha=\sum_{i=1}^n g_i$. Since $\alpha g_j=g_j\alpha=\alpha$ for all $j\in\{1,\dots,n\}$, the set $I=K[G]\alpha$ is an ideal of $K[G]$. Since, moreover,
$$\alpha^2=\sum_{i=1}^n g_i\alpha=n\alpha=0$$
in the field K, it follows that I is a nilpotent non-zero ideal. Thus $J(K[G])\neq\{0\}$, as Proposition 9.6 yields $I\subseteq J(K[G])$. □

Since the Jacobson radical of a group algebra of a finite group contains every nil left ideal, the following consequence of the theorem follows immediately:

30.2. Corollary. *Let G be a finite group. Then $K[G]$ does not contain non-zero nil left ideals.*

§ 31. Herstein's theorem

Our aim now is to answer the following question: When a group algebra is algebraic? Herstein's theorem provides a solution in the case of fields of characteristic zero. In prime characteristic, the problem is still open.

31.1. Definition. A group G is **locally finite** if every finitely generated subgroup of G is finite.

If G is a locally finite group, then every element $g\in G$ has finite order, as the subgroup $\langle g\rangle$ is finite because it is finitely generated.

31.2. Example. Every finite group is locally finite

31.3. Example. The group $\mathbb{Z}$ is not locally finite because it is torsion-free.

31.4. Example. Let p be a prime number. The **Prüfer's group**
$$\mathbb{Z}(p^\infty)=\{z\in\mathbb{C}:z^{p^n}=1\text{ for some }n\in\mathbb{Z}_{>0}\},$$
is locally finite.

31.5. Example. Let X be an infinite set and $\mathbb{S}_X$ be the set of bijective maps $X \to X$ moving only finitely many elements of X. Then $\mathbb{S}_X$ is locally finite.

A group G is a **torsion** group if every element of G has finite order. Locally finite groups are torsion groups.

31.6. Example. Abelian torsion groups are locally finite. Let G be a locally finite abelian group and H be a finitely generated subgroup. Since G is an abelian torsion group, so is H. Thus H is finite by the structure theorem of abelian groups.

31.7. Proposition. *Let G be a group and N be a normal subgroup of G. If N and G/N are locally finite, then G is locally finite.*

Proof. Let $\pi\colon G \to G/N$ be the canonical map and $\{g_1, \dots, g_n\}$ be a finite subset of G. Since G/N is locally finite, the subgroup Q of G/N generated by $\pi(g_1), \dots, \pi(g_n)$ is finite, say

$$Q = \{\pi(g_1), \dots, \pi(g_n), \pi(g_{n+1}), \dots, \pi(g_m)\}$$

for some $g_{n+1}, \dots, g_m \in G$.

For each $i, j \in \{1, \dots, n\}$ there exist $u_{ij} \in N$ and $k \in \{1, \dots, m\}$ such that

$$g_i g_j = u_{ij} g_k.$$

Let U be the subgroup of N generated by $\{u_{ij} : 1 \leq i, j \leq n\}$. Since N is locally finite, U is finite. Moreover, since each $g_i g_j g_l$ can be written as

$$g_i g_j g_l = u_{ij} g_k g_l = u_{ij} u_{kl} g_t = u g_t$$

for some $u \in U$ and $t \in \{1, \dots, m\}$, it follows that the subgroup H of G generated by $\{g_1, \dots, g_n\}$ is finite, as $|H| \leq m|U|$. $\square$

31.8. Bonus exercise. Prove that solvable torsion groups are locally finite.

31.9. Theorem (Herstein). *If G is a locally finite group, then $K[G]$ is algebraic. Conversely, if $K[G]$ is algebraic and K has characteristic zero, then G is locally finite.*

Proof. Assume that G is locally finite. Let $\alpha \in K[G]$. The subgroup $H = \langle \operatorname{supp} \alpha \rangle$ is finite, as it is finitely generated. Since $\alpha \in K[H]$ and $\dim_K K[H] < \infty$, the set $\{1, \alpha, \alpha^2, \dots\}$ is linearly dependent. Thus α is algebraic over K.

Let $\{x_1, \dots, x_m\}$ be a finite subset of G. Adding inverses if needed, we may assume that $\{x_1, \dots, x_m\}$ generates the subgroup $H = \langle x_1, \dots, x_m \rangle$ as a semigroup. Let

$$\alpha = x_1 + \cdots + x_m \in K[G].$$

Since α is algebraic over K, there exist $b_0, b_1, \dots, b_{n+1} \in K$ such that

$$b_0 + b_1\alpha + \cdots + b_{n+1}\alpha^{n+1} = 0,$$

where $b_{n+1} \neq 0$. We can rewrite this as

$$\alpha^{n+1} = a_0 + a_1\alpha + \cdots + a_n\alpha^n$$

for some $a_0, \dots, a_n \in K$. Note that

$$\alpha^k = (x_1 + \cdots + x_m)^k = \sum x_{i_1} \cdots x_{i_k}$$

for all k. Two words $x_{i_1} \cdots x_{i_k}$ and $x_{j_1} \cdots x_{j_l}$ could represent the same element of the group H. In this case, the coefficient of $x_{i_1} \cdots x_{i_k} = x_{j_1} \cdots x_{j_l}$ in α^k will be a positive integer ≥ 2.

Let $w = x_{i_1} \cdots x_{i_{n+1}} \in H$ be a word of length $n+1$. Since K is of characteristic zero, it follows that $w \in \operatorname{supp}(\alpha^{n+1})$. Since, moreover, $\alpha^{n+1} = \sum_{j=0}^{n} a_j \alpha^j$, it follows that $w \in \operatorname{supp}(\alpha^j)$ for some $j \in \{0, \dots, n\}$. Thus each word in the letters x_j of length $n+1$ can be written as a word in the letters x_j of length $\leq n$. Therefore H is finite and hence G is locally finite. $\square$

§ 32. Formanek's theorem, I

We start with some exercises.

32.1. Exercise. Let K be a field. Let A be a K-algebra algebraic over K and $a \in A$.

1) a is a left zero divisor if and only if a is a right zero divisor.
2) a is left invertible if and only if a is right invertible.
3) a is invertible if and only if a is not a zero divisor.

32.2. Exercise. For $\alpha = \sum_{g \in G} \alpha_g g \in \mathbb{C}[G]$ let $|\alpha| = \sum_{g \in G} |\alpha_g| \in \mathbb{R}$. Prove the following statements:

1) $|\alpha + \beta| \leq |\alpha| + |\beta|$, and
2) $|\alpha\beta| \leq |\alpha||\beta|$

for all $\alpha, \beta \in \mathbb{C}[G]$.

32.3. Theorem (Formanek). *Let G be a group. If every element of $\mathbb{Q}[G]$ is invertible or a zero divisor, then G is locally finite.*

Proof. Let $\{x_1, \dots, x_n\}$ be a finite subset of G. Adding inverses if needed, we may assume that $\{x_1, \dots, x_n\}$ generates the subgroup $H = \langle x_1, \dots, x_n \rangle$ as a semigroup. Let

$$\alpha = \frac{1}{2n}(x_1 + \cdots + x_n) \in \mathbb{Q}[G]$$

Note that $|\alpha| \leq 1/2$. We claim that $1 - \alpha \in \mathbb{Q}[G]$ is invertible. If not, then it is a zero divisor. If there exists $\delta \in \mathbb{Q}[G]$ such that $\delta(1-\alpha) = 0$, then $\delta = \delta\alpha$. Since

$$|\delta| = |\delta\alpha| \leq |\delta||\alpha| \leq |\delta|/2,$$

it follows that $\delta = 0$. Similarly, $(1-\alpha)\delta = 0$ implies $\delta = 0$.

Let $\beta = (1-\alpha)^{-1} \in \mathbb{Q}[G]$. For each k let

$$\gamma_k = (1 + \alpha + \cdots + \alpha^k) - \beta.$$

Then

$$\begin{aligned}\gamma_k(1-\alpha) &= (1 + \alpha + \cdots + \alpha^k - \beta)(1-\alpha) \\ &= (1 + \alpha + \cdots + \alpha^k)(1-\alpha) - \beta(1-\alpha) = -\alpha^{k+1}\end{aligned}$$

and thus $\gamma_k = -\alpha^{k+1}\beta$. Since

$$|\gamma_k| = |-\alpha^{k+1}\beta| \leq |\beta||\alpha^{k+1}| \leq \frac{|\beta|}{2^{k+1}},$$

it follows that $\lim_{k\to\infty} |\gamma_k| = 0$.

We now prove that $H \subseteq \operatorname{supp}\beta$. This will finish the proof of the theorem, as $\operatorname{supp}\beta$ is a finite subset of G by definition. If $H \not\subseteq \operatorname{supp}\beta$, let $h \in H \setminus \operatorname{supp}\beta$. Assume that

$h = x_{i_1} \cdots x_{i_m}$ is a word in the letters x_j of length m. Let c_j be the coefficient of h in α^j. Then $c_0 + \cdots + c_k$ is the coefficient of h in γ_k, but

$$|\gamma_k| \geq c_0 + c_1 + \cdots + c_k \geq c_m > 0$$

for all $k \geq m$, as each c_j is non-negative, a contradiction to $|\gamma_k| \to 0$ if $k \to \infty$. $\square$

§ 33. Tensor products

The **tensor product** of the vector spaces (over K) U and V is the quotient vector space $K[U \times V]/T$, where $K[U \times V]$ is the vector space with basis

$$\{(u, v) : u \in U, v \in V\}$$

and T is the subspace generated by elements of the form

$$(\lambda u + \mu u', v) - \lambda(u, v) - \mu(u', v), \quad (u, \lambda v + \mu v') - \lambda(u, v) - \mu(u, v')$$

for $\lambda, \mu \in K$, $u, u' \in U$ and $v, v' \in V$. The tensor product of U and V will be denoted by $U \otimes_K V$ or $U \otimes V$ when the base field is clear from the context. For $u \in U$ and $v \in V$ we write $u \otimes v$ to denote the coset $(u, v) + T$.

33.1. Theorem. *Let U and V be vector spaces. Then there exists a bilinear map*

$$U \times V \to U \otimes V, \quad (u, v) \mapsto u \otimes v,$$

such that each element of $U \otimes V$ is a finite sum of the form

$$\sum_{i=1}^{N} u_i \otimes v_i$$

for some $u_1, \dots, u_N \in U$ and $v_1, \dots, v_N \in V$. Moreover, if W is a vector space and

$$\beta \colon U \times V \to W$$

is a bilinear map, there exists a linear map $\overline{\beta} \colon U \otimes V \to W$ such that $\overline{\beta}(u \otimes v) = \beta(u, v)$ for all $u \in U$ and $v \in V$.

Proof. By definition, the map

$$U \times V \to U \otimes V, \quad (u, v) \mapsto u \otimes v,$$

is bilinear. From the definitions, it follows that $U \otimes V$ is a finite linear combination of elements of the form $u \otimes v$, where $u \in U$ and $v \in V$. Since $\lambda(u \otimes v) = (\lambda u) \otimes v$ for all $\lambda \in K$, the first claim follows.

Since the elements of $U \times V$ form a basis of $K[U \times V]$, there exists a linear map

$$\gamma \colon K[U \times V] \to W, \quad \gamma(u, v) = \beta(u, v).$$

Since β is bilinear by assumption, $T \subseteq \ker \gamma$. It follows that there exists a linear map $\overline{\beta} \colon U \otimes V \to W$ such that

$$\begin{tikzcd} K[U \times V] \arrow[r] \arrow[d] & W \\ U \otimes V \arrow[ur, dashed] & \end{tikzcd}$$

commutes. In particular, $\overline{\beta}(u \otimes v) = \beta(u, v)$. $\square$

33.2. Exercise. Prove that the properties of the previous theorem characterize tensor products up to isomorphism.

Some properties:

33.3. Proposition. *Let $\varphi\colon U\to U_1$ and $\psi\colon V\to V_1$ be linear maps. There exists a unique linear map $\varphi\otimes\psi\colon U\otimes V\to U_1\otimes V_1$ such that*

$$(\varphi\otimes\psi)(u\otimes v)=\varphi(u)\otimes\psi(v)$$

for all $u\in U$ and $v\in V$.

Proof. Since $U\times V\to U_1\otimes V_1$, $(u,v)\mapsto\varphi(u)\otimes\psi(v)$, is bilinear, there exists a linear map $U\otimes V\to U_1\otimes V_1$, $u\otimes v\to\varphi(u)\otimes\psi(v)$. Thus

$$\sum u_i\otimes v_i\mapsto\sum\varphi(u_i)\otimes\psi(v_i)$$

is well-defined. $\square$

33.4. Exercise. Prove the following statements:

1) $(\varphi\otimes\psi)(\varphi'\otimes\psi')=(\varphi\varphi')\otimes(\psi\psi')$.
2) If φ and ψ are isomorphisms, then $\varphi\otimes\psi$ is an isomorphism.
3) $(\lambda\varphi+\lambda'\varphi')\otimes\psi=\lambda\varphi\otimes\psi+\lambda'\varphi'\otimes\psi$.
4) $\varphi\otimes(\lambda\psi+\lambda'\psi')=\lambda\varphi\otimes\psi+\lambda'\varphi\otimes\psi'$.
5) If $U\simeq U_1$ and $V\simeq V_1$, then $U\otimes V\simeq U_1\otimes V_1$.

The following proposition is extremely useful:

33.5. Proposition. *If U and V are vector spaces, then $U\otimes V\simeq V\otimes U$.*

Proof. Since $U\times V\to V\otimes U$, $(u,v)\mapsto v\otimes u$, is bilinear, there exists a linear map

$$U\otimes V\to V\otimes U,\quad u\otimes v\mapsto v\otimes u.$$

Similarly, there exists a linear map

$$V\otimes U\to U\otimes V,\quad v\otimes u\mapsto u\otimes v.$$

Thus $U\otimes V\simeq V\otimes U$. $\square$

33.6. Exercise. Prove that $(U\otimes V)\otimes W\simeq U\otimes(V\otimes W)$.

33.7. Exercise. Prove that $U\otimes K\simeq U\simeq K\otimes U$.

33.8. Proposition. *Let U and V be vector spaces. If $\{u_1,\dots,u_n\}$ is a linearly independent subset of U and $v_1,\dots,v_n\in V$ is such that $\sum_{i=1}^n u_i\otimes v_i=0$, then $v_i=0$ for all $i\in\{1,\dots,n\}$.*

Proof. Let $i\in\{1,\dots,n\}$ and

$$f_i\colon U\to K,\quad f_i(u_j)=\delta_{ij}=\begin{cases}1 & \text{if } i=j,\\ 0 & \text{otherwise.}\end{cases}$$

Since the map

$$U\times V\to V,\quad (u,v)\mapsto f_i(u)v,$$

is bilinear, there exists a linear map $\alpha_i\colon U\otimes V\to V$ such that $\alpha_i(u\otimes v)=f_i(u)v$. Thus

$$v_i=\sum_{j=1}^n \alpha_i(u_j\otimes v_j)=\alpha_i\left(\sum_{j=1}^n u_j\otimes v_j\right)=0. \qquad \square$$

33.9. EXERCISE. Prove that $u\otimes v=0$ and $v\neq 0$ imply $u=0$.

33.10. THEOREM. *Let U and V be vector spaces. If $\{u_i : i\in I\}$ is a basis of U and $\{v_j : j\in J\}$ is a basis of V, then $\{u_i\otimes v_j : i\in I, j\in J\}$ is a basis of $U\otimes V$.*

PROOF. The $u_i\otimes v_j$ are generators of $U\otimes V$, as $u=\sum_i \lambda_i u_i$ and $v=\sum_j \mu_j v_j$ imply

$$u\otimes v=\sum_{i,j}\lambda_i\mu_j u_i\otimes v_j.$$

We now prove that the $u_i\otimes v_j$ are linearly independent. We need to show that each finite subset of the $u_i\otimes v_j$ is linearly independent. If $\sum_k\sum_l \lambda_{kl}u_{i_k}\otimes v_{j_l}=0$, then

$$0=\sum_k u_{i_k}\otimes\left(\sum_l \lambda_{kl}v_{j_l}\right).$$

Since the u_{i_k} are linearly independent, Proposition 33.8 implies that $\sum_l \lambda_{kl}v_{j_l}=0$. Thus $\lambda_{kl}=0$ for all k,l, as the v_{j_l} are linearly independent. $\square$

If U and V are finite-dimensional vector spaces, then

$$\dim(U\otimes V)=(\dim U)(\dim V).$$

33.11. COROLLARY. *If $\{u_i : i\in I\}$ is a basis of U, then every element of $U\otimes V$ can be written uniquely as a finite sum $\sum_i u_i\otimes v_i$.*

PROOF. Every element of the tensor product $U\otimes V$ is a finite sum of the form $\sum_i x_i\otimes y_i$, where $x_i\in U$ and $y_i\in V$. If $x_i=\sum_j \lambda_{ij}u_j$, then

$$\sum_i x_i\otimes y_i=\sum_i\left(\sum_j \lambda_{ij}u_j\right)\otimes y_i=\sum_j u_j\otimes\left(\sum_i \lambda_{ij}y_i\right). \qquad \square$$

33.12. EXERCISE. Let A and B be algebras. Prove that $A\otimes B$ is an algebra with

$$(a\otimes b)(x\otimes y)=ax\otimes by.$$

33.13. EXERCISE. Let K be a field and A,B,C be K-algebras. Prove the following statements:

1) $A\otimes B\simeq B\otimes A$.
2) $(A\otimes B)\otimes C\simeq A\otimes(B\otimes C)$.
3) $A\otimes K\simeq A\simeq K\otimes A$.
4) If $A\simeq A_1$ and $B\simeq B_1$, then $A\otimes B\simeq A_1\otimes B_1$.

Some examples:

33.14. PROPOSITION. *If G and H are groups, then $K[G]\otimes K[H]\simeq K[G\times H]$.*

Proof. The set $\{g \otimes h : g \in G, h \in H\}$ is a basis of $K[G] \otimes K[H]$ and the elements of $G \times H$ form a basis of $K[G \times H]$. There exists a linear isomorphism

$$K[G] \otimes K[H] \to K[G \times H], \quad g \otimes h \mapsto (g, h),$$

that is multiplicative. Thus $K[G] \otimes K[H] \simeq K[G \times H]$ as algebras. □

33.15. Proposition. *If A is an algebra, then $A \otimes K[X] \simeq A[X]$.*

Proof. Each element of the tensor product $A \otimes K[X]$ can be written uniquely as a finite sum of the form $\sum a_i \otimes X^i$. Routine calculations show that $A \otimes K[X] \mapsto A[X]$, $\sum a_i \otimes X^i \mapsto \sum a_i X^i$, is a linear algebra isomorphism. □

33.16. Exercise. Prove that if A is an algebra, then $A \otimes M_n(K) \simeq M_n(A)$. In particular, $M_n(K) \otimes M_m(K) \simeq M_{nm}(K)$.

Proposition 33.15 and Exercise 33.16 are examples of a procedure known as **scalar extensions**.

33.17. Theorem. *Let A be an algebra over K and E be an extension of K (this just simply means that K is a subfield of E). Then $A^E = E \otimes_K A$ is an algebra over E with respect to the scalar multiplication*

$$\lambda(\mu \otimes a) = (\lambda\mu) \otimes a,$$

for all $\lambda, \mu \in E$ and $a \in A$.

Proof. Let $\lambda \in E$. Since $E \times A \to E \otimes_K A$, $(\mu, a) \mapsto (\lambda\mu) \otimes a$, is K-bilinear, there exists a linear map $E \otimes_K A \to E \otimes_K A$, $\mu \otimes a \mapsto (\lambda\mu) \otimes a$. The scalar multiplication is then well-defined and

$$\lambda(u + v) = \lambda u + \lambda v$$

for all $\lambda \in E$ and $u, v \in E \otimes_K A$. Moreover,

$$(\lambda + \mu)u = \lambda u + \mu u, \quad (\lambda\mu)u = \lambda(\mu u), \quad \lambda(uv) = (\lambda u)v = u(\lambda v)$$

for all $u, v \in E \otimes_K A$ and $\lambda, \mu \in E$. □

33.18. Exercise. Prove the following statements:

1) $\{1\} \otimes A$ is a subalgebra of A^E isomorphic to A.

2) If $\{a_i : i \in I\}$ is a basis of A, then $\{1 \otimes a_i : i \in I\}$ is a basis of A^E.

33.19. Exercise. Prove that if G is a group and K is a subfield of E, then

$$E \otimes_K K[G] \simeq E[G].$$

§ 34. Formanek's theorem, II

The combination of technique known as extensions of scalars we have seen in the previous section and Formanek's theorem for rational group algebras yield the following general result.

34.1. Theorem (Formanek). *Let K be a field of characteristic zero and let G be a group. If every element of $K[G]$ is invertible or a zero divisor, then G is locally finite.*

Proof. Since K is of characteristic zero, $\mathbb{Q} \subseteq K$. Then $K[G] \simeq K \otimes_{\mathbb{Q}} \mathbb{Q}[G]$. Note that each $\beta \in K \otimes_{\mathbb{Q}} \mathbb{Q}[Q]$ can be written uniquely as

$$\beta = 1 \otimes \beta_0 + \sum k_i \otimes \beta_i,$$

where $\{1, k_1, k_2, \dots,\}$ is a basis of K as a $\mathbb{Q}$-vector space. Let $\alpha \in \mathbb{Q}[G] \subseteq K[G]$. By assumption, every element of $K[G]$ is invertible or a zero divisor. Assume first that α is invertible. Let $\beta \in K[G] \simeq K \otimes_{\mathbb{Q}} \mathbb{Q}[Q]$ be such that $\alpha\beta = 1$. Since

$$1 \otimes 1 = (1 \otimes \alpha)\beta = 1 \otimes \alpha\beta_0 + \sum k_i \otimes \alpha\beta_i,$$

it follows that $\alpha\beta_0 = 1$. Thus α is invertible in $\mathbb{Q}[G]$. Similarly, if α is a zero divisor, then $\alpha\beta = 0$ and $\alpha\beta_j = 0$ for all j. Thus α is a zero divisor in $\mathbb{Q}[G]$. Hence each $\alpha \in \mathbb{Q}[G]$ is invertible or a zero divisor, so Formanek's theorem for $\mathbb{Q}$ applies. $\square$

§ 35. Wedderburn's little theorem

35.1. Definition. The n-th cyclotomic polynomial is defined as the polynomial

$$\Phi_n(X) = \prod (X - \zeta), \tag{35.1}$$

where the product is taken over all n-th primitive roots of unity.

Some examples:

$$\begin{aligned}
\Phi_2 &= X - 1,\\
\Phi_3 &= X^2 + X + 1,\\
\Phi_4 &= X^2 + 1,\\
\Phi_5 &= X^4 + X^3 + X^2 + X + 1,\\
\Phi_6 &= X^2 - X + 1,\\
\Phi_7 &= X^6 + X^5 + \cdots + X + 1.
\end{aligned}$$

35.2. Lemma. *If $n \in \mathbb{Z}_{>0}$, then*

$$X^n - 1 = \prod_{d|n} \Phi_d(X).$$

Proof. Write

$$X^n - 1 = \prod_{j=1}^{n} (X - e^{2\pi ij/n}) = \prod_{d|n} \prod_{\substack{1 \leq j \leq n \\ \gcd(j,n)=d}} (X - e^{2\pi ij/n}) = \prod_{d|n} \Phi_d(X). \qquad \square$$

35.3. Lemma. *If $n \in \mathbb{Z}_{>0}$, then $\Phi_n(X) \in \mathbb{Z}[X]$.*

Proof. We proceed by induction on n. The case $n = 1$ is trivial, as $\Phi_1(X) = X - 1$. Assume that $\Phi_d(X) \in \mathbb{Z}[X]$ for all $d < n$. Then

$$\prod_{d|n, d \neq n} \Phi_d(X) \in \mathbb{Z}[X]$$

is a monic polynomial. Thus $\Phi_n(X)/\prod_{d|n,d<n} \Phi_d(X) \in \mathbb{Z}[X]$. $\square$

35.4. Exercise. Let a,n,m be positive integers. Prove that

$$\gcd(a^n-1,a^m-1)=a^{\gcd(n,m)}-1.$$

35.5. Theorem (Wedderburn). *Every finite division ring is a field.*

Proof. Let D be a finite division ring and $K=Z(D)$. Then K is a finite field, say $|K|=q$. We claim that $|q-\zeta|>q-1$ for all n-th root of one $\zeta\neq 1$. In fact, write $\zeta=\cos\theta+i\sin\theta$. Then $\cos\theta<1$ and

$$|q-\zeta|^2=q^2-(2\cos\theta)q+1>(q-1)^2.$$

Note that D is a K-vector space. Let $n=\dim_K D$. We claim that $n=1$. If $n>1$, the class equation for the group $D^\times=D\setminus\{0\}$ implies that

$$q^n-1=q-1+\sum_{j=1}^{m}\frac{q^n-1}{q^{d_j}-1}, \tag{35.2}$$

where $1<\frac{q^n-1}{q^{d_j}-1}\in\mathbb{Z}$ for all $j\in\{1,\dots,m\}$. In fact, each centralizer, including the zero element, is a vector space over K and hence has q^{d_j} elements for some d_j. By the fundamental counting principle, the orbit has size $(q^n-1)/(q^{d_j}-1)$.

Since $d^{d_j}-1$ divides q^n-1 (see Exercise 35.4), each d_j divides n. In particular, (35.1) implies that

$$X^n-1=\Phi_n(X)(X^{d_j}-1)h(X) \tag{35.3}$$

for some $h(X)\in\mathbb{Z}[X]$. By evaluating (35.3) in $X=q$ we obtain that $\Phi_n(q)$ divides q^n-1 and that $\Phi_n(q)$ divides $\frac{q^n-1}{q^{d_j}-1}$. By (35.2), $\Phi_n(q)$ divides $q-1$. Thus

$$q-1\geq|\Phi_n(q)|=\prod|q-\zeta|>q-1,$$

as each $|q-\zeta|>q-1$, a contradiction. $\square$

There are several proofs of Wedderburn's theorem. For example, [**25**] contains a proof that uses only elementary linear algebra. In [**21**, Chapter 14] the theorem is proved using group theory.

35.6. Theorem. *Let D be a division ring of characteristic $p>0$. If G is a subgroup of $D\setminus\{0\}$, then G is cyclic.*

We shall need a lemma. The lemma uses a well-known result from elementary number theory: If φ is the Euler function that counts the positive integers up to a given integer n that are relatively prime to n, then

$$\sum_{d\mid n}\varphi(d)=n.$$

Let us present a quick group-theoretical proof. Let $G=\langle g\rangle$ be the cyclic group of order n. Then

$$n=|G|=\sum_{1\leq d\leq n}|\{g\in G:|g|=d\}|=\sum_{d\mid n}|\{g\in G:|g|=d\}|$$

by Lagrange's Theorem. Since G is cyclic, for each $d\mid n$, $\langle g^{n/d}\rangle$ is the unique subgroup of G of order d. Now the claim follows, as each subgroup of the form $\langle g^{n/d}\rangle$ has $\varphi(d)$ generators.

35.7. Lemma. *Let K be a field. Any finite subgroup of $K \setminus \{0\}$ is cyclic.*

Proof. Let G be a finite subgroup of $K \setminus \{0\}$ and $n = |G|$. For a divisor d of n, let $f(d)$ be the number of elements of G of order d. Then

$$\sum_{d|n} f(d) = n. \tag{35.4}$$

We claim that if $d \mid n$ is such that $f(d) \neq 0$, then $f(d) = \varphi(d)$, where φ is the Euler function. In fact, if $f(d) \neq 0$, then there exists $g \in G$ such that $|g| = d$. Let $H = \langle g \rangle$ be the subgroup of G generated by g. Every element of H is a root of the polynomial $p(X) = X^d - 1 \in K[X]$. Since $p(X)$ has at most d roots, H is the set of roots of $p(X)$. In particular, $g^m \in H$ and $|g^m| = d$ if and only if $\gcd(m, d) = 1$. Hence $f(d) = \varphi(d)$.

Since $\sum_{d|n} \varphi(d)$ and (35.4), it follows that $f(n) = \varphi(n) \neq 0$. Hence there exists $g \in G$ such that $|g| = n = |G|$ and G is cyclic. $\square$

Proof of Theorem 35.6. Let $F = \sum_{g \in G} (\mathbb{Z}/p)g$. Then F is a finite subring of D. Since D is a domain, F is a domain. Let $\alpha \in F \setminus \{0\}$. Then $\{\lambda\alpha : \lambda \in F\} = F$. Since $\lambda\alpha = 1$ for some $\lambda \in F$, F is a division ring. By Wedderburn's theorem, F is a field. Note that $G \subseteq F$. Therefore G is cyclic by the previous lemma. $\square$

§ 36. Zsigmondy's theorem

One of Wedderburn's original proof of Theorem 35.5 uses a result proved by Zsigmondy [**27**]. Zsigmondy's theorem is quite popular in mathematical contests.

36.1. Theorem (Zsigmondy). *Let $a > b \geq 1$ be such that $\gcd(a, b) = 1$ and $n \geq 2$. Then there exists a prime divisor of $a^n - b^n$ that does not divide $a^k - b^k$ for all $k \in \{1, \dots, n-1\}$ except when $n = 2$ and $a + b$ is a power of two or $(a, b, n) = (2, 1, 6)$.*

Proof. See for example [**26**]. $\square$

We now quickly sketch a proof of Wedderburn's theorem 35.5 based on Zsigmondy's theorem.

Let D be a division ring of dimension n over $\mathbb{Z}/p$ for a prime number p. Assume first that there exists a prime number q such that $q \nmid p$ and the order of p modulo q is n. Let $x \in D \setminus \{0\}$ be an element of order q and F be the subring of D generated by g. Note that F is a finite-dimensional $(\mathbb{Z}/p)$-vector space. Let $m = \dim F$. Since $g^{p^m - 1} = 1$, q divides $p^m - 1$. Thus $m = n$ and hence $D = F$ is commutative.

Assume now that there is no prime number q such that $q \nmid p$ and the order of p modulo q is n. By Zsigmondy's theorem, $n = 2$ or $n = 6$ and $p = 2$. If $n = 2$, then D is commutative, as it is the subring generated by any element of $D \setminus \mathbb{Z}/p$. If $n = 6$ and $p = 2$, then the order of 2 modulo 9 is 6. Since $D \setminus \{0\}$ contains a subgroup of order 9 and all groups of order 9 are abelian, we can use the previous argument to complete the proof.

§ 37. Fermat's last theorem in finite rings

37.1. Theorem. *Let K be a finite field and A be a finite-dimensional K-algebra. For $n \geq 1$, there exist $x, y, z \in A \setminus \{0\}$ such that $x^n + y^n = z^n$ if and only if A is not a division algebra.*

Proof. Assume first that A is a division algebra. By Wedderburn's theorem, A is a finite field, say $|A| = q$. Then $x^{q-1} = 1$ for all $x \in A \setminus \{0\}$. Hence $x^n + y^n = z^n$ does not have a solution.

Conversely, assume that A is not a division algebra. In particular, A is not a field and $|A| > 2$. The equation $x + y = z$ has a solution in $A \setminus \{0\}$ (for example, $x = 1$, $y = z - 1$ and $z \notin \{0, 1\}$ is a solution). Since $\dim A < \infty$, the Jacobson radical $J(A)$ is nilpotent. There are two cases to consider.

If $J(A) \neq \{0\}$, then there exists $a \in A \setminus \{0\}$ such that $a^2 = 0$. Thus $a^n = 0$ for all $n \geq 2$. Hence $x^n + y^n = z^n$ has a non-trivial solution in $A \setminus \{0\}$ for all $n \geq 2$ (for example, take $x = a$ and $y = z = 1$).

If $J(A) = \{0\}$, then A is semisimple and $A \simeq \prod_{i=1}^k M_{n_i}(D_i)$ for (finite) division rings $D_1, \dots, D_k$ and integers $n_1, \dots, n_k$. By Wedderburn's theorem, each D_i is a finite field. We consider two possible cases.

If there exists $i \in \{1, \dots, k\}$ such that $n_i > 1$, then $M_{n_i}(D_i)$ has non-zero elements such that their squares are zero. Thus there exists $x \in A \setminus \{0\}$ such that $x^2 = 0$. In particular, $x^n + y^n = z^n$ has a solution.

If $k \geq 2$, then $x = (1, 0, 0, \dots, 0)$, $y = (0, 1, 0, \dots, 0)$ and $z = (1, 1, 0, \dots, 0)$ is a solution of $x^n + y^n = z^n$. $\square$

§ 38. Frobenius's theorem

38.1. Theorem (Frobenius). *Every finite-dimensional real division algebra is isomorphic to $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$.*

We present an elementary proof. We shall need some lemmas.

38.2. Lemma. *Let D be a real division algebra such that $\dim D = n$. If $x \in D$, then there exists $\lambda \in \mathbb{R}$ such that $x^2 + \lambda x \in \mathbb{R}$.*

Proof. Since $\dim D = n$, the set $\{1, x, x^2, \dots, x^n\}$ is linearly dependent. So there exists a non-zero polynomial $f(X) \in \mathbb{R}[X]$ of degree $\leq n$ such that $f(x) = 0$. Without loss of generality, we may assume that the leading coefficient of $f(X)$ is one. Then we can write $f(X)$ as a product of polynomials of degree ≤ 2, say

$$f(X) = (X - \alpha_1) \cdots (X - \alpha_r)(X^2 + \lambda_1 X + \mu_1) \cdots (X^2 + \lambda_s X + \mu_s).$$

Since D is a division algebra and $f(x) = 0$, some factor of $f(X)$ is zero at x. If $x - \lambda_j \neq 0$ for all j, then x is a root of some $X^2 + \lambda_k X + \mu_k$. In any case, there exists $\lambda \in \mathbb{R}$ such that $x^2 + \lambda x \in \mathbb{R}$. $\square$

38.3. Lemma. *Let D be a real division algebra of dimension n. Then*

$$V = \{x \in D : x^2 \in \mathbb{R}_{\leq 0}\}$$

is a subspace of D such that $D = \mathbb{R} \oplus V$.

Proof. If $x \in D \setminus V$ is such that $x^2 \in \mathbb{R}$, then, since $x^2 > 0$, it follows that $x^2 = \alpha^2$ for some $\alpha \in \mathbb{R}$. Thus $x = \pm\alpha \in \mathbb{R}$, as D is a division algebra and

$$(x - \alpha)(x + \alpha) = x^2 - \alpha^2 = 0.$$

We claim that V is a subspace of D. Note that $0 \in V$ and that if $x \in V$, then $\lambda x \in V$ for all $\lambda \in \mathbb{R}$. Let $x, y \in V$. If $\{x, y\}$ is linearly dependent, then $x + y \in V$. If not, we claim that $\{1,x,y\}$ is linearly independent. If there exist $\alpha,\beta,\gamma\in\mathbb{R}$ such that $\alpha x+\beta y+\gamma=0$, then

$$\alpha^2x^2=\beta^2y^2+2\beta\gamma y+\gamma^2=(-\beta y-\gamma)^2.$$

This implies that $2\beta\gamma y\in\mathbb{R}$ and thus $\beta\gamma=0$, as $0\leq 4\beta^2\gamma^2y^2\leq 0$. Hence $\alpha=\beta=\gamma=0$. (If $\beta=0$, then $0\leq\gamma^2=\alpha^2x^2\leq 0$. If $\gamma=0$, then $\alpha=\beta=0$ because $\{x,y\}$ is linearly independent.) The previous lemma implies that there exist $\lambda,\mu\in\mathbb{R}$ such that

$$(x+y)^2+\lambda(x+y)\in\mathbb{R},\quad (x-y)^2+\mu(x-y)\in\mathbb{R}.$$

Since

$$(x+y)^2+(x-y)^2=2x^2+2y^2\in\mathbb{R},$$

it follows that $(\lambda+\mu)x+(\lambda-\mu)y\in\mathbb{R}$. Since $\{1,x,y\}$ is linearly independent, $\lambda=\mu=0$. Thus $(x+y)^2\in\mathbb{R}$. If $x+y\notin V$, then, the first paragraph of the proof implies that $x+y\in\mathbb{R}$, a contradiction.

Clearly, $\mathbb{R}\cap V=\{0\}$. If $x\in D\setminus\mathbb{R}$, then the previous lemma implies that $x^2+\lambda x\in\mathbb{R}$ for some $\lambda\in\mathbb{R}$. We claim that $x+\lambda/2\in V$. If not, since

$$(x+\lambda/2)^2=x^2+\lambda x+(\lambda/2)^2\in\mathbb{R},$$

it follows that $x+\lambda/2\in\mathbb{R}$ and thus $x\in\mathbb{R}$, a contradiction. Hence

$$x=-\lambda/2+(x+\lambda/2)\in\mathbb{R}\oplus V. \qquad\square$$

38.4. Lemma. *Let D be a real division algebra of (real) dimension n. If $n>2$, then there exist $i,j,k\in D$ such that $\{1,i,j,k\}$ is linearly independent and*

$$i^2=j^2=k^2=-1,\qquad ij=-ji=k,\qquad ki=-ik=j,\qquad jk=-kj=i. \tag{38.1}$$

Proof. Let $V=\{x\in D:x^2\in\mathbb{R},x^2\leq 0\}$ be the subspace of Lemma 38.3. For $x,y\in V$ let $x*y=xy+yx=(x+y)^2-x^2-y^2\in\mathbb{R}$. If $x\neq 0$, then $x*x=2x^2\neq 0$. Since $\dim V=n-1$, there exist $y,z\in V$ such that $\{y,z\}$ is linearly independent. Let

$$x=z-\frac{z*y}{y*y}y.$$

Since $\{y,z\}$ is linearly independent, $x\neq 0$. Moreover, since

$$x*y=\left(z-\frac{z*y}{y*y}\right)*y=zy-\frac{z*y}{y*y}y^2+yz-\frac{z*y}{y*y}y^2=z*y-\frac{z*y}{y*y}y*y=0,$$

it follows that $xy=-yx$. Let

$$i=\frac{1}{\sqrt{-x^2}}x,\quad j=\frac{1}{\sqrt{-y^2}}y,\quad k=ij.$$

A direct calculation shows that the formulas of (38.1) hold. For example,

$$ji=\frac{1}{\sqrt{-y^2}}\frac{1}{\sqrt{-x^2}}yx=\frac{1}{\sqrt{-x^2}}\frac{1}{\sqrt{-y^2}}(-xy)=-k. \qquad\square$$

Now we are finally ready to prove the theorem:

Proof of 38.1. Let D be a real division algebra and let $n=\dim D$. If $n=1$, then $D\simeq\mathbb{R}$. If $n=2$, the subspace V of Lemma 38.3 is non-zero and thus there exists $i\in D$ such that $i^2=-1$. Hence $D\simeq\mathbb{C}$. Lemma 38.4 implies that $n\neq 3$. If $n=4$, then $D\simeq\mathbb{H}$. Suppose that $n>4$. By Lemma 38.4 there exist $i,j,k\in D$ such that $\{1,i,j,k\}$ is linearly independent and that the formulas of (38.1) hold. Let

$$V=\{x\in D:x^2\in\mathbb{R}_{\leq 0}\}.$$

By Lemma 38.3, $\dim V=n-1$. Thus there exists $x\in V\setminus\langle i,j,k\rangle$. Let

$$e=x+\frac{i*x}{2}i+\frac{j*x}{2}j+\frac{k*x}{2}k\in V\setminus\{0\}.$$

A direct calculation shows that $i*e=j*e=k*e=0$. Then

$$ek=e(ij)=(ei)j=-(ie)j=-i(ej)=i(je)=(ij)e=ke,$$

a contradiction. □

§ 39. Jacobson's commutativity theorem

We start with an easy exercise.

39.1. Exercise. A ring R is **boolean** if $x^2=x$ for all $x\in R$. Prove that boolean rings are commutative.

To prove this fact, note that $1=(-1)^2=-1$. This means that R has characteristic two. Let $x,y\in R$. Since $x+y=(x+y)^2=x^2+xy+yx+y^2$. it follows that $0=xy+yx$ and hence $xy=yx$.

39.2. Exercise. Let $n\geq 1$ and R be a ring such that $x^{3\cdot 2^n}=x$ for all $x\in R$. Prove that R is commutative.

Hint: Note that R has characteristic two and that $x^{2^{n+1}}=x^{2^n}$ for all $x\in R$. Thus R is a boolean ring.

39.3. Proposition. *Let R be a finite ring such that for each $x\in R$ there exists $n(x)\geq 2$ such that $x^{n(x)}=x$. Then R is commutative.*

Proof. Since R is finite, R is artinian and hence $J(R)$ is nil. Since R is reduced, $J(R)=\{0\}$. By the Artin–Wedderburn theorem, $R\simeq\prod_{i=1}^k M_{n_i}(D_i)$ for some division rings $D_1,\dots,D_k$. Since R is finite, each D_i is finite. By Wedderburn's theorem, every D_i is a field. Again, since R is reduced, $n_i=1$ for all i. Therefore R is commutative, as it is a direct product of finitely many fields. □

In this lecture, we will extend the result of Proposition 39.3 to arbitrary (i.e. non-finite) rings.

39.4. Theorem (Jacobson). *Let R be a ring such that for each $x\in R$ there exists $n(x)\geq 2$ such that $x^{n(x)}=x$. Then R is commutative.*

We shall need the following lemma.

39.5. Lemma. *Let K be a finite field of characteristic $p>0$. There exists $n\in\mathbb{Z}_{>0}$ such that $|K|=p^n$ and $x^{p^n}=x$ for all $x\in K$. Moreover, if $K\setminus\{0\}=\{x_1,\dots,x_{p^n-1}\}$, then $X^{p^n}-X=(X-x_1)\cdots(X-x_{p^n-1})X$.*

Proof. The field K is a $(\mathbb{Z}/p)$-vector space. If $\dim_{\mathbb{Z}/p} K = n$, then $|K| = p^n$. In particular, $K \setminus \{0\}$ is an abelian group of order $p^n - 1$ and hence, by Lagrange's theorem, $x^{p^n-1} = 1$ for all $x \in K \setminus \{0\}$. Thus $x^{p^n} = x$ for all $x \in K$ and hence every $x \in K$ is a root of the polynomial $X^{p^n} - X$ of degree p^n. □

Let R be a ring. For each $r \in R$ the map $\operatorname{ad} r \colon R \to R$, $x \mapsto rx - xr$, is a derivation. This means that $\operatorname{ad}(xy) = (\operatorname{ad} x)y + x(\operatorname{ad} y)$ for all $x, y \in R$. By induction one proves that

$$(\operatorname{ad} r)^n(x) = \sum_{k=0}^{n} (-1)^k \binom{n}{k} r^{n-k} x r^k \tag{39.1}$$

for all $x \in R$ and $n \in \mathbb{Z}_{>0}$. If p is a prime number, p divides $\binom{p}{k}$ for all $k \in \{1, \dots, p-1\}$. This fact is needed to solve the following exercise:

39.6. Exercise. Let p be a prime number and R be a ring of characteristic p. Prove that $(\operatorname{ad} r)^{p^n} = \operatorname{ad} r^{p^n}$.

Now we are ready to prove Jacobson's commutativity theorem.

Proof of Theorem 39.4. We divide the proof into several steps and claims. We may assume that R is non-zero.

Claim. $J(R) = \{0\}$.

Let $x \in J(R)$ and $n = n(x)$. Since $-x^{n-1} \in J(R)$, there exists $y \in R$ such that $-x^{n-1} \circ y = -x^{n-1} + y - x^{n-1}y = 0$. Thus

$$-x^{n-1} + y = x^{n-1}y \implies -x + xy = x(-x^{n-1} + y) = x^n y = xy.$$

This implies that $x = 0$.

Claim. Without loss of generality, we may assume that R is primitive.

Let $\{P_i : i \in I\}$ be the collection of primitive ideals of R. The map $R \to \prod_{i \in I} R/P_i$, $r \mapsto (r + P_i)_{i \in I}$, is an injective homomorphism, since its kernel is

$$\bigcap_{i \in I} P_i = J(R) = \{0\}.$$

Note that R is commutative if each R/P_i is commutative. Moreover, each R/P_i satisfies the assumption, that is $(x + P_i)^{n(x)} = x^{n(x)} + P_i = x + P_i$, and and is a primitive ring.

Claim. R is a division ring.

By Jacobson's density theorem, there exists a division ring D and a D-vector space V such that R is dense in V. We claim that $\dim_D V = 1$. If $\dim_D V \geq 2$, let $\{v_1, v_2\} \subseteq V$ be a linearly independent set. Then there exists $f \in R$ such that $f(v_1) = v_2$ and $f(v_2) = 0$. This implies that $f^k(v_1) = 0$ for all $k \geq 2$ and $f(v_1) \neq 0$. This contradicts the fact that $f^n = f$ for $n = n(f)$. Thus $R \simeq D^{\mathrm{op}}$, a division ring.

Claim. R has positive characteristic.

Since R is a division ring, $2 = 1 + 1 \in R$. There exists $n \geq 2$ such that $2^n = 2$. In particular, $2(2^{n-1} - 1) = 0$. This implies the claim.

Claim. Every non-zero subring of R is a division ring.

Let $S \subseteq R$ is a non-zero subring of R. If $x \in S \setminus \{0\}$, then $x^{n(x)} = x$. In particular,

$$x^{-1} = x^{n(x)-2} \in S.$$

Claim. R is commutative.

Let us assume that R is not commutative. Let $x \in R \setminus Z(R)$. Since R has positive characteristic, there exists $m > 0$ such that $mx = 0$. Moreover, since R is a division ring and $x^{n(x)} = x$, it follows that $x^{n(x)-1} = 1$. These facts imply that the subring K of R generated by x is finite. By Wedderburn's theorem, K is a finite field. Thus $|K| = p^k$ for some prime number p and some $k > 0$ and

$$x^{p^k} = x.$$

Note that R is a K-vector space and $\delta = \operatorname{ad} x \colon R \to R$, $y \mapsto xy - yx$, is a K-linear map. Moreover, by the exercise,

$$\delta^{p^k} = (\operatorname{ad} x)^{p^k} = \operatorname{ad}\left(x^{p^k}\right) = \operatorname{ad} x = \delta$$

and

$$\delta(\delta - x_1 \operatorname{id}) \cdots (\delta - x_{p^{k-1}} \operatorname{id}) = 0 \tag{39.2}$$

if $K = \{0, x_1, \dots, x_{p^k-1}\}$. Since x is not central, δ is non-zero. So there exists $y \in R$ such that $\delta(y) \neq 0$. Evaluating (39.2) in y and using that R is a division ring, we obtain that

$$x_i y = \delta(y) = xy - yx$$

for some i. Let R_0 be the subring of R generated by x and y. Since $xy - yx = \delta(y) \neq 0$, the ring R_0 is a non-commutative division ring. Note that $yx = (x - x_i)y \in Ky$, as $x \in K$ and $x_i \in K$. By induction one proves that $yx^j \subseteq Ky$ for all $j \geq 1$ and hence $y^i K \subseteq Ky^i$ for all $i \geq 1$. This implies that

$$K + Ky + \cdots + Ky^{n(y)-2} \subseteq R$$

is a subring. It follows that $K + Ky + \cdots + Ky^{n(y)-2} = R_0$, as it is a subring of R included in R_0 that contains x and y. Since R_0 is a finite division ring, it is a field by Wedderburn's theorem, a contradiction since it is non-commutative. □

There are elementary proofs of Jacobson's commutativity theorem. See for example [**18**].

List of projects

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